Islands on Land

From Centralization to Redistribution:

A territorial strategy for equal service access in rural northern Sweden.



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Islands on Land
From Centralization to Redistribution

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Abstract

Islands on land examines how urban bias shapes everyday life in rural northern Sweden. Using Holmön, an island in Umeå Municipality, as a case study, the project investigates how transport dependency, reduced public presence, and the withdrawal of welfare services affect the possibility of living there. Because of its isolated geography, Holmön serves as a geographically intensified case of peripheral rurality, concentrating systemic conditions that are also present across other rural settlements.

In response, the thesis develops Islands on Land as a territorial framework that rethinks Umeå Municipality as an archipelago of semi-autonomous islands regardless of geographic location or population. Based on this framework, a speculative redistribution strategy is made to redistribute essential services such as healthcare and primary schools according to both population and distance, in order to test a different spatial logic of access. The strategy also takes into architectural scale through the Samhällsrum at Prästgården, which combines health, school, and community functions in response to the spatial demands of service redistribution. The project also identifies existing typologies for reuse across the municipality where the Samhällsrum concept can be extended.

Key Words: Rural, Redistribution, Territorial plannign, Welfare

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Introduction

Urbanity has gradually become the default setting for contemporary architecture and planning, where metropolitan conditions are often treated as the norm and everything else as deviation. In Sweden, this bias is reinforced by a major historical shift in settlement patterns: In just over 200 years, Sweden has shifted from a largely rural society to one in which nearly 88 percent of the population lives in urban areas (SCB, 2020). As urbanity has come to stand for development and success, rural places are increasingly assessed through systems that do not fully match their geography, population patterns or everyday needs.

This thesis examines how urban bias shapes rural livability in northern Sweden, using Holmön in Umeå Municipality as its case study. I first came to Holmön through the Umeå School of Architecture summer course, where we designed and built a sauna for the island. During those two weeks, I got a glimpse of the small but resilient local community that helps sustain Holmön through its own effort and investment. That experience made clear how closely everyday life on the island depends on local cooperation and on the continued willingness of residents to compensate for the withdrawal of public support. Holmön's isolation makes the consequences of infrastructural withdrawal and centralized service provision especially visible, while also revealing the forms of adaptability and resilience that emerge in response.

The thesis asks what it means to live in a place where access to essential services is shaped by distance, transport dependency and a weak public presence. More specifically, it investigates how conditions of shrinkage and isolation can be understood not just as problems to be managed, but as starting points for rethinking rural planning.

In response, the thesis develops Islands on Land, a speculative redistribution strategy that rethinks Umeå Municipality as an archipelago of semi-autonomous islands regardless of geographic location or population. Through this framework, essential services such as healthcare and primary schools are redistributed according to both population and distance, in order to test a different spatial logic of access. The project then translates that strategy into architectural form through the Samhällsrum at Prästgården, where redistributed services are given a local spatial implementation. It also identifies existing typologies for reuse across the municipality where the Samhällsrum concept can be extended.

Aim and Research Questions

How can Islands on Land challenge urban-centric welfare models through the territorial redistribution of essential services in support of rural life in northern Sweden?

The aim of this thesis is to explore how the territorial redistribution of essential services can support rural livability and challenge urban-centric welfare models in northern Sweden.

Scope and Delimitations

This thesis is limited to northern Sweden, with Umeå Municipality and Holmön as its primary territorial focus. It concentrates on essential services, particularly healthcare and primary schooling, as these are central to rural livability and public presence. The project is developed as a speculative architectural and territorial study, meaning that it explores alternative spatial logics rather than proposing a complete policy or implementation model.

The thesis does not attempt to solve the wider structural conditions of rural decline, nor does it provide a full administrative or economic framework for service redistribution. Instead, it tests how architectural thinking can reveal territorial inequalities and suggest spatial strategies that may support rural life more broadly.

Method

This thesis uses a research-by-design methodology combining literature review, case study analysis, mapping, and speculative design. Theoretical reading provides the conceptual framework, while Holmön serves as the case through which territorial conditions are examined. Municipal documents, previous reports, and workshop material from Med ön som katalysator step 1 and 2 form the empirical basis for the analysis.

The spatial part of the method develops through mapping and the Islands on Land framework, which tests alternative service distributions across Umeå Municipality. This is then translated into a building-scale proposal through the Samhällsrum at Prästgården. The method moves between analysis and design in order to test how territorial redistribution can be understood and represented architecturally.

Part 1

Defining Rural and the Welfare Contract

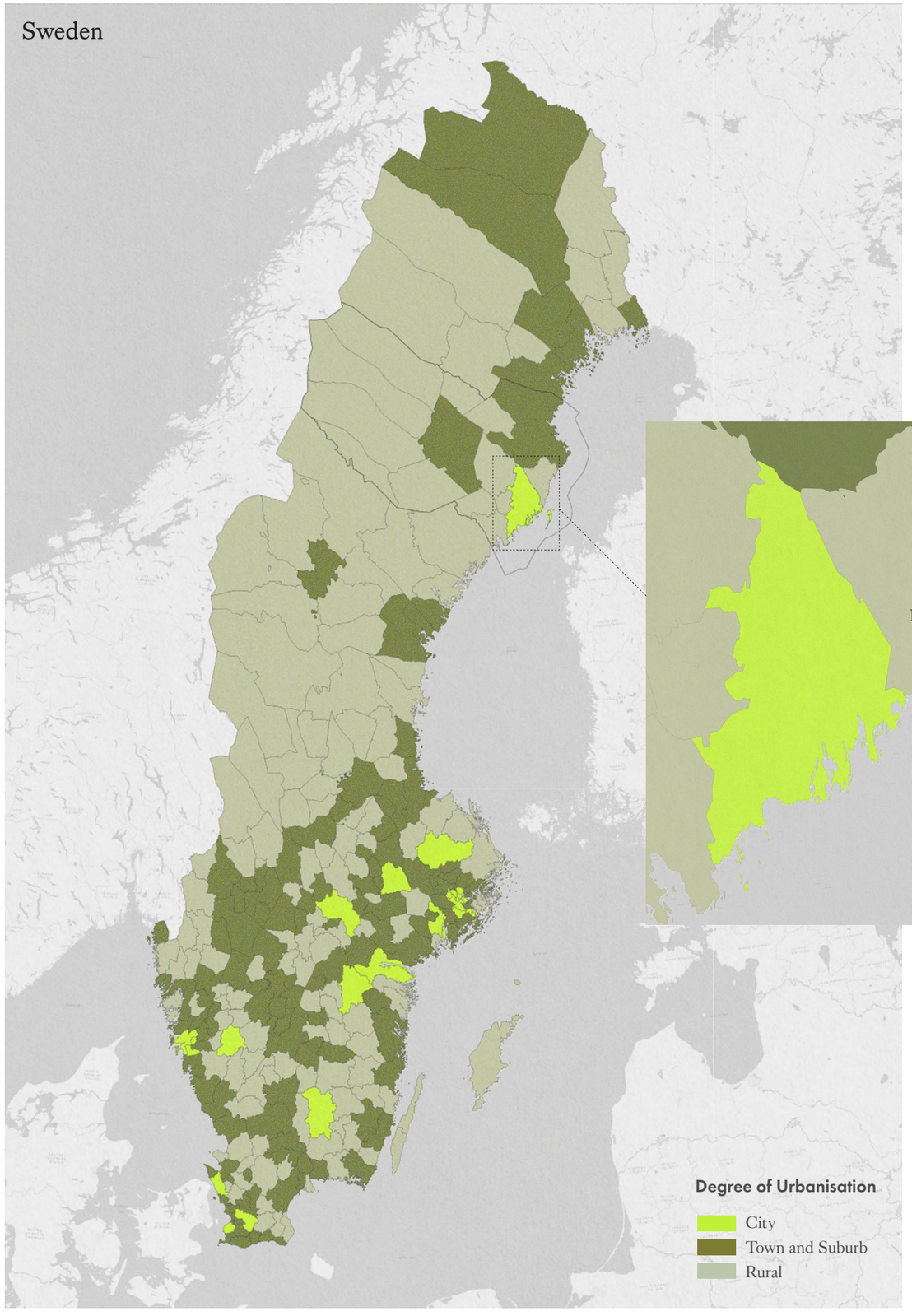
Urban Norms and Centralization

Urbanity has gradually become the default setting for contemporary architecture and planning. This “urban norm” has reached a point where theory, practice, and education frequently treat metropolitan conditions as the standard, while all other spatialities are framed as deviations. In the Swedish context, this bias is reinforced by a dramatic historical shift in settlement patterns; today, approximately 88 percent of the population resides in tätorter, a stark contrast to just over 200 years ago when roughly 90 percent lived in the countryside (SCB 2020).

This accelerating urbanization and centralization significantly alter the societal view of architecture and housing development. Urbanity has become the spatial symbol for development, growth, and success(Åkerman, 2020. p). This creates a self-fulfilling cycle where growth is only recognized in dense clusters, leaving smaller towns to struggle for survival. As early as 1977, Christopher Alexander noted that the big city acts as a “magnet,” making it “terribly hard for small towns to stay alive and healthy in the face of central urban growth” (p. 34). This tension produces increasing spatial and economic inequalities between the city and the hinterland.

Rural areas are not static spaces of idyllic or traditional living, but landscapes that are constantly transformed through economic, social and environmental processes. In Inget landskap är en ö: Dialektik och praktik i öländska landskap, Saltzman examines contemporary rural environments on Öland to show how landscapes are formed, lived in and negotiated, and argues that ideas of preservation often rest on a selective nostalgia that overlooks the fact that these places are always in motion rather than frozen in time (Saltzman, 2001 p. 12-8; 238-239). This tension between a desire to “keep” a landscape and the reality of continuous change is highly relevant for rural territories more broadly, especially when rural is defined residually as everything outside urban areas and marketed through images of proximity to nature and an idealized countryside. In the Swedish context, the culturally familiar image of the red cottage with white trim by a field, forest or lake condenses a powerful romantic idea of the countryside that many people feel attached to, even if they have never lived that life themselves. Such nostalgic attachments can encourage a view of rural landscapes as stable and self-contained, when in fact they are tightly entangled with wider dynamics such as commuting, tourism, resource extraction, climate change and global markets. Thinking with Saltzman’s argument, it becomes important to recognize that rural landscapes do not exist in a bubble; they are shaped by decisions, flows, and conflicts that extend far beyond the village. And regular people are often more affected by this because of the lack of public presence.

In architecture and planning, rurality must be understood as a field of architectural agency in order to challenge the urban-centric bias of the discipline. AMO and OMA’s Countryside, A Report (2020), presented through the Solomon R. Guggenheim Museum as both exhibition and publication, signals this shift by treating the countryside as a major architectural subject. As a project by one of architecture’s most influential offices, it gives rural territory disciplinary visibility and weight. This is important because the question of rurality is not only a regional or municipal issue, but a disciplinary and societal one as well.



Definition of Rural

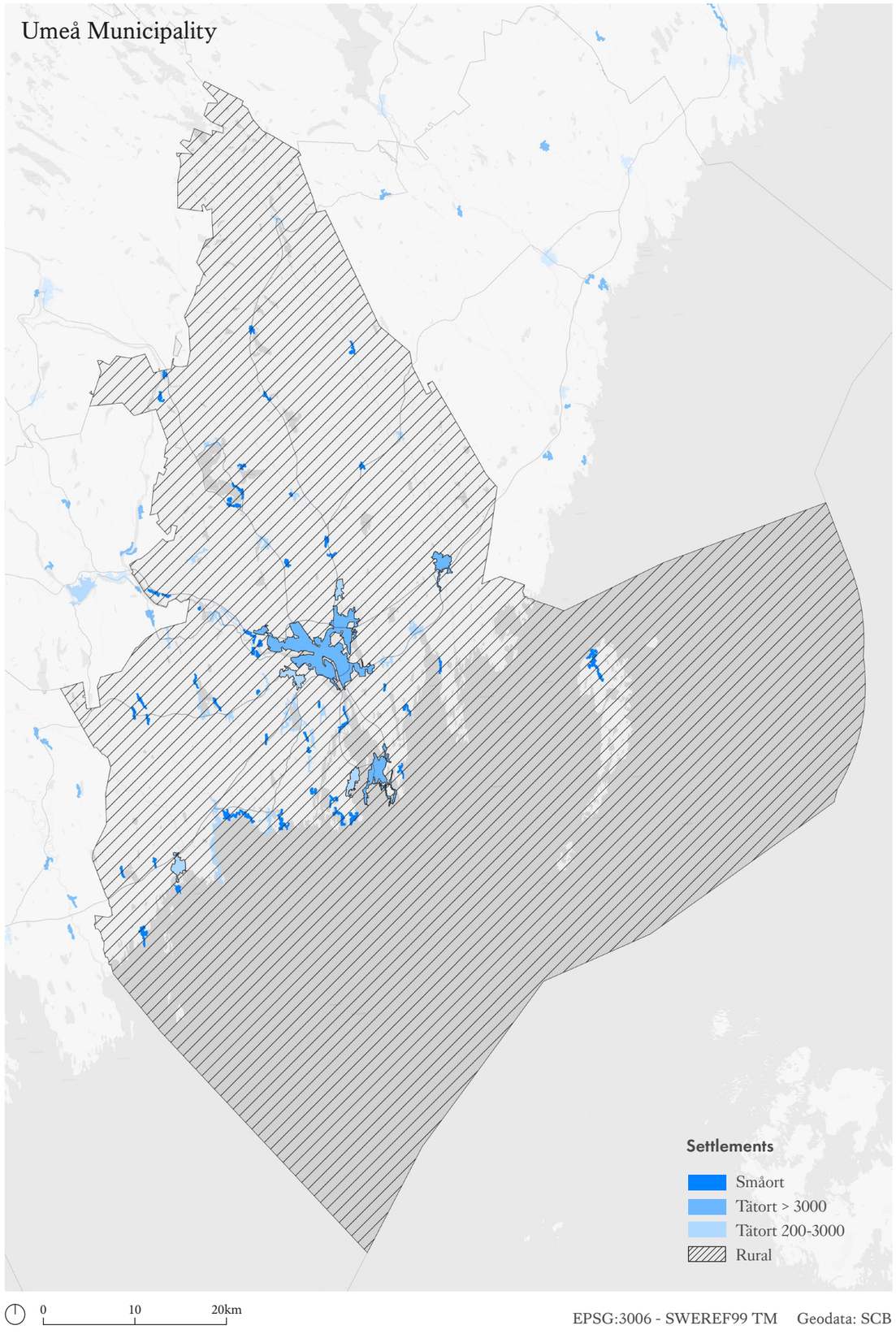
The definition of rural might seem obvious, but it is not consistent between countries or even within them. In Sweden, different authorities use parallel classification systems depending on their mandate and the type of statistics or planning decisions they support (Westholm, 2008, p. 49–58), and in most cases, many refrain from having a specific definition. Westholm argues that there is no need for a single, uniform categorization of rural areas and warns that rigid definitions risk becoming a “straitjacket” in a time when statistical tools can describe settlement patterns in very fine detail (Westholm, 2008). This flexibility is valuable in a large country like Sweden, where rural conditions differ between the north and south, the inland and the coast, but it also raises questions about how vague notions of the countryside translate into planning and concrete action.

Statistics Sweden (SCB) does not define landsbygd but instead defines tätort as any settlement with more than 200 residents, and villages with fewer than 200 residents are further classified as småort (SCB 2020, p. 4). Glesbygd is an area located more than a 45-minute drive from the nearest tätort with more than 3,000 inhabitants, as well as islands without fixed connections to the mainland (Glesbygdsverket, 2007).

Many classification systems rely on Local Administrative Units (LAUs) or municipality-wide population densities. For example, the European Commission’s Degree of Urbanisation (DEGURBA) system classifies an entire municipality as a “city” if over 50 percent of its population lives in high-density grid cells of at least 1,500 inhabitants/km² (Eurostat, 2025). In the classification systems that use LAUs, Umeå Municipality is classified as a “city” (urban) Municipality (Eurostat, 2025). However, this high-level categorization creates a mismatch between statistical data and lived reality. Because Umeå municipality’s population is concentrated to its urban core, peripheral settlements like Holmön and Botsmark are statistically absorbed into urban classifications. These areas are “urban” on paper, but they remain peripheral in terms of investment, infrastructure representation, and political attention.

The lack of precision is a specific challenge in the Nordic context. As Nordregio (2024) points out, Nordic countries generally have much larger local administrative units than the rest of Europe, which makes LAU-based systems like DEGURBA particularly blunt instruments for our geography. To oppose this, Nordregio has proposed a new grid-based territorial typology that looks at 1 km² cells across the entire Nordic region (Nordregio, 2024). By shifting the focus from administrative borders to a more precise grid, it becomes possible to reveal the diversity of the landscape. Highlighting that when we rely on outdated administrative categories, we reinforce the urban-rural divide by failing to see, and therefore failing to represent.

Umeå Municipality defines its rural territory as everything outside of the major tätorter: Umeå, Röbäck, Obbola, Holmsund, and Sävar (Umeå Municipality 2022, p. 4). By this definition, Sävar, a tätort with roughly 3,000 inhabitants located around 20 kilometers from Umeå, is not classified as rural according to the municipal definition. However, despite its ”urban” status, Sävar faces several challenges typically associated with rural areas (Bergvall Kalén 2026). This creates a secondary layer of classification friction: a settlement can be large enough to not be rural on a map, yet remain ”Rural” in its functional struggles.



Definitions of Rural Umeå Municipality

DEGURBA
City *(UN, OECD, and European Union)*

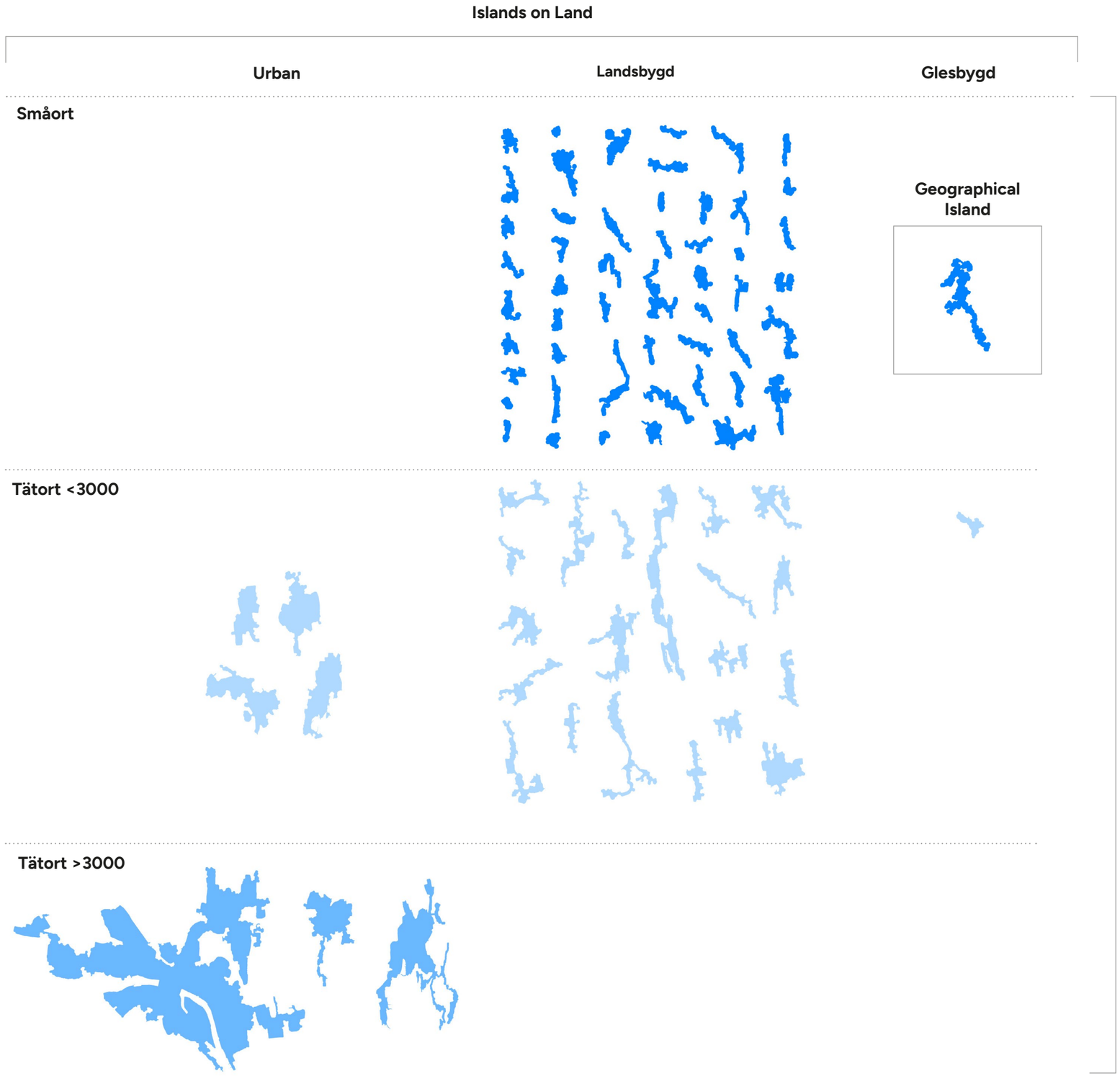
Småort
A settlement with less than 200 inhabitants *(SCB)*

Tätort
a settlement with at least 200 inhabitants where buildings are normally no more than 200 meters apart *(SCB)*

Glesbygd
Areas located more than a 45-minute drive from the nearest urban area (tätort) with more than 3,000 inhabitants, as well as islands without fixed mainland connections 45 minutes *(Glesbygdsverket)*

Umeå Municiaplity Rural Definition
Umeå Municipality defines the rural area as all parts of the municipality except for the major urban areas of Umeå (including Ersmark and Röbäck), Sävar, Holmsund, Obbola, and Hörnefors. *(Umeå Municipality 2021)*

Islands on Land
All settlements are defined as Islands on Land, with the borders of each island set by SCB's outlines of tätorter and småorter. The definition is used to examine autonomy, access to essential services, and transport dependence.



According to Jennie Bergvall Kalén, rural development strategist at Umeå Municipality, this official boundary serves a practical purpose. By excluding larger settlements like Sävar from the "rural" category, the municipality's rural efforts can prioritize resources to more exposed villages, those characterized by long distance from urban centers and a weak public presence (Bergvall Kalén 2026).

Dahlberg, Kohl and Liang (2024) analyze geographic income inequality in Sweden (1991–2017) by testing different rural-urban definitions to identify where gaps are largest and thus where policy action could have most impact. The article study compares incomes across degrees of urbanization and finds the biggest differences between municipality types rather than within them, demonstrating how the choice of rural definition determines the scale and location of identified challenges, with implications for targeting rural development and investments (Dahlberg, Kohl and Liang, 2024). Further highlighting Westholm's argument of the need for flexibility of the definitions in relation to usage (Westholm, 2008).

The Swedish Welfare Contract

Both the Nordic and the Swedish welfare models are built on a social contract where the state assumes collective responsibility for key needs such as education, healthcare and social care, and where public authorities at local, regional and national level are expected to secure access to core services for everyone (Nordiskt samarbete, 2025; Regeringskansliet, 2017). This commitment has been central to how welfare states justify taxation and redistribution, and to how citizens understand their rights and obligations in relation to the state.

At the same time, research and official reports indicate that these ambitions are increasingly difficult to uphold in many rural and sparsely populated areas. Studies of glesbygd and landsbygd highlight a long-term trend of school and preschool closures, difficulties in recruiting qualified staff, lower healthcare funding for the regions that have been in deficit since 2022 (SCB 2025), and elderly care services are further apart and fewer in relation to the number of elderly. Which means that residents in smaller towns and villages often face longer travel distances and reduced local access to basic welfare functions, particularly in northern Sweden, where settlements are far apart, and infrastructure is thin (Skolverket 2025; Myndigheten för vård- och omsorgsanalys 2021).

Architecture and Welfare
Scandinavian Perspectives shows how the Nordic welfare state has been materially produced through architecture, landscape and planning, from housing and schools to recreational landscapes and infrastructures, and how these environments have given spatial form to ideas of equality, care and a "good life" for all citizens (Arrhenius, Braae and Ruud, 2024). The Swedish Folkhemmet emerges in this history as a paradigmatic example of how politics and architecture once worked together to deliver far-reaching social reforms, where state-led housing programs, social infrastructure and everyday environments were deliberately designed to universalize welfare standards. Read in relation to the current situation of rural shrinkage, welfare cuts and the social contract that promised equivalent service points to a need for a new alliance between politics and spatial design.

Part 2

Reading the Current Planning System Through Holmön

How can understanding the lived conditions on Holmön reveal an urban bias in regional and national development strategies?

This thesis uses Holmön as a case to examine how the current planning system affects the island and how it distributes welfare services across the municipality. Holmön is a small island in northern Sweden, a part of the country that is already underrepresented in national debates, and its geography often leaves it outside standard planning categories. As a shrinking and aging community that is both rural and an island, it makes the consequences of an urban-centered planning norm especially visible. The preliminary study ”Med ön som katalysator” (Bergvall Kalén and Otterbjörk, 2025) describes Holmön’s challenges as part of a broader regional pattern of rural service access.Building on that study’s attention to service provision on Holmön, this thesis uses services as a lens to examine how planning, welfare, and everyday livability are connected in rural places.

Historical Context

Holmön is a small island community in the Kvarken strait, within Umeå Municipality, located 45 minutes by ferry from the mainland (Trafikverket, 2026). Much of Holmön is of national interest for nature conservation and outdoor recreation, and traces of habitation date back to the 1300s. In statistical terms, the settlement is classified as a småort with a population of less than 200 but more than 50 people (SCB). It is also defined as glesbygd without a permanent connection to the mainland. In the early twentieth century, the island had around 450 residents, compared to roughly 60 today, with 39 living there year-round (Holmöns Utvecklingsforum, 2020; Holmöns Utvecklingsforum, 2000). Historically, Holmön was sustained by local livelihoods in fishing, forestry, and small-scale agriculture. Over the twentieth century, these activities became increasingly unprofitable as production was industrialized within a growth-driven economy (Holmöns Utvecklingsforum, 2000). As these livelihoods disappeared, the permanent population declined, and the remaining community has grown older. Summer tourism and events such as Visfestivalen now bring a seasonal influx of visitors, but this does not create the stable work, services, or infrastructure needed for year-round living.

In 2013, Umeå Municipality explicitly stated that it should be possible to live, work, and commute daily to and from the island(Kommunstyrelsen Umeå Kommun, 2013). More than a decade later, this major goal remains unachieved. A primary barrier is the ferry situation. Today, Trafikverket manages the line using old ferries that cannot safely moor in strong northern winds, alongside a strict ice directive that grounds vessels during freezing periods. When the infrastructure fails, hovercraft or delayed weekday-only helicopters take over, creating a vulnerability that can leave residents isolated for up to eight days (Alexandersson and Bergvall Kalén, 2025).
However, this is not the only reason for not being able to live on Holmön year-round, while the physical failure of the ferry is the most visible barrier, the inability to sustain year-round residency on Holmön is equally driven by the systematic withdrawal of social infrastructure.

The challenges on Holmön are not new phenomena. Archival records show that the same discourse regarding population decline, school closures, and tourism as a potential salvation has been looping for over fifty years. Local coverage from the 1960s and 1970s already questioned whether the island would ”die out” and become a winter vacuum used only for summer recreation (Västerbottens-Kuriren, 1964; Västerbottens Folkblad, 1978, material accessed through holmon.info, compiled by Sven Nordblad.). By the late 1980s, public broadcast discussions echoed the same sentiment, framing unreliable transport as a barrier to local industry and positioning tourism as the only viable option (SVT, 1987) This half-century of identical discourse makes it clear that Holmön’s challenges are rooted in a structural problem with historical roots.

Systemic Challenges on Holmön

The preliminary report *Med ön som katalysator*, produced by Umeå municipality, Region Västerbotten and Holmöns Utvecklingsforum, proposes local actions on Holmön that could serve as a model for other Nordic rural areas. The report identifies several systemic challenges present on the island but also across the region. Challenges such as a lack of flexibility and local knowledge, unclear roles and social contract, transportation-dependent systems, centralization, urban norms, and a lack of regional goals. While Holmön’s physical isolation and amplified isolation due to the transportation issues are extreme. Its struggles mirror those of other peripheral settlements, such as mountain villages that experience high seasonal strain on infrastructure like water and waste management. By treating Holmön as an example, it becomes clear that its ”exceptional” status reflects a planning culture that lacks established routines for anything that falls outside the urban standard. The report frames Holmön’s challenges also described as regionally systemic rather than isolated issues. In my designproject i have in more depth looked at 4 of these challenges, all in relation to welfare services.

Centralization Logic and Urban Norms

“The lack of flexibility and local knowledge within public administration, together with the high degree of transportation dependency, can also be linked to the fact that today’s service system is largely centralized. It is based on a logic that favors densely populated areas over more sparsely populated ones.” (Bergvall Kalén and Otterbjörk, 2025, p. 17)

Locally, this plays out through welfare systems and operational models that are designed around urban conditions and then applied as if they were universally valid. Service structures, efficiency measures and minimum standards calibrated for larger, denser cities drive centralization and withdrawal when imposed on sparsely populated areas. In practice, this dismantling of welfare in rural areas is closely tied to a lack of flexibility in how national goals are implemented, as rural municipalities and regions are expected to deliver “likvärdig service” within frameworks built for cities. Making it increasingly difficult to sustain everyday life and to attract or retain residents outside the major urban centers.

The centralization of welfare and governance is not only an abstract norm but is also clearly visible in the way Västerbotten and Umeå have been reorganized over time.

At the regional scale, municipal reforms in the 1900s transformed a landscape of small municipalities derived from church parishes into a smaller number of larger units. Before the 1952 reform, the municipal division committee found that many of the municipalities consisted of less than 500 inhabitants and due to the municipalities' increased responsibility, it was decided to merge many of them together and in a later reform in 1964 it was decided that municipalities also needed a functioning central locality to operate efficiently (SCB, 2023). Västerbotten used to consist of many small rural municipalities, and the region's three larger towns, Umeå, Lycksele and Skellefteå, formed their own separate city municipalities (SCB 1951). Holmön was also an independent municipality and, before it was merged with Umeå in 1974, Sweden's smallest municipality, with only 175 inhabitants (HUF, 2000). The successive reforms reduced Västerbotten to 15 municipalities, and in this process smaller rural entities, including Holmön, were absorbed into larger administrative structures.

At the municipal scale, Umeå has undergone a similar internal centralization. In 2014, the municipality reorganized its political structure and phased out the kommundelsnämnder, the local district boards (Mitz, 2014). These boards were critiqued for having low engagement and limited influence, but their removal, rather than reinforcing peripheral settlements, further centralized decision-making to the municipal core and weakened formal arenas for local representation in outlying areas. Together, these shifts at the regional and municipal level show how political and administrative centralization has gradually redefined rural and peripheral places as parts of larger units, while also reducing their institutional visibility.

Lack of public presence

"The lack of public presence is perceived as just as crucial to being able to live on the island year-round as the lack of functioning transport. The absence of a continuous public presence contributes to residents feeling cut off from the welfare system." (Bergvall Kalén and Otterbjörk, 2025, p. 16)

The absence of continuous public presence on Holmön leaves residents feeling disconnected from the welfare system. Access to public services is not only a practical question of getting care or education, but also central to feeling recognized and supported as a citizen, and it is therefore vital for sustaining year-round living (Bergvall Kalén and Otterbjörk, 2025, p. 16). Institutions such as schools and healthcare facilities act as visible markers of public presence; when they are gone, both geographic and political isolation deepen, and transportation constraints further restrict access to services that are intended to be universally available (Nordiskt samarbete, 2025). The withdrawal of public actors shifts responsibility onto the local community, where a small number of people carry much of the work of keeping the island livable, often without formal recognition (Bergvall Kalén and Otterbjörk, 2025, p. 16)

The closure of Holmön's school in 2009 is a key example of this withdrawal (Sveriges Radio, 2009). In the Swedish welfare model, education is a mandatory right (skolplikt), yet school viability is currently assessed using urban-density standards. This creates a paradox for families: they are legally required to ensure their children attend school, but the municipality does not provide a local facility, and the transport-dependent alternative is often unreliable due to weather (SVT Nyheter, 2009). Families with children are thus structurally excluded from Holmön, not because they do not want to live there, but because they cannot guarantee that they can meet their legal obligations under existing service and transport arrangements.



Västerbotten 1952

1. Nordmaling
2. Hörnefors
3. Bjurholm
4. Degerfors
5. Vännäs
6. Vännäs Köping
7. Holmsunds köping
8. Umeå
9. Umeå Landskommun
10. Sävar
11. Holmön
12. Bygdeå
13. Nysätra
14. Lövånger
15. Burträsk
16. Bureå
17. Skellefteå
18. Skellefteå Landskommun
19. Byske
20. Jörn
21. Norsjö
22. Malå
23. Örträsk
24. Lycksele
25. Lyksele Landskommun
26. Stensele
27. Sorsele
28. Tärna
29. Vilhelmina
30. Vilhelmina Köping
31. Åsele
32. Fredrika

Västerbotten Today

1. Bjurholm
2. Dorotea
3. Lycksele
4. Malå
5. Nordmaling
6. Norsjö
7. Robertsfors
8. Skellefteå
9. Sorsele
10. Storuman
11. Umeå
12. Vilhelmina
13. Vindeln
14. Vännäs
15. Åsele

Lack of flexibility and local knowledge

"The lack of public presence contributes to a knowledge gap around Holmön and the particular conditions that exist here — knowledge that is crucial to the public sector's ability to respond to actual needs." (Bergvall Kalén and Otterbjörk, 2025, p. 16)

The preliminary report also identifies a lack of flexibility and local knowledge. Standardized operational models, designed for larger and denser contexts, leave little room to adapt services to the island's specific conditions (Bergvall Kalén and Otterbjörk, 2025, p. 16). Even when local organizations such as Holmöns Utvecklingsforum propose community-driven childcare solutions, they are rejected because they do not meet "urban norm" requirements for number of children (Eriksson, 2023). In practice, the system treats deviation from the norm as grounds for denial rather than as a reason to adjust the model.

For older residents, the same rigidity is visible in elder care. As healthcare and elderly care services are centralized to larger tätorter, the "care gap" in sparsely populated areas grows, and existing operational models are not designed to reach island-like settlements (Bergvall Kalén, 2026). Umeå Municipality itself notes that deficient traffic to Holmön makes it difficult for the municipality to fulfill parts of its assignment, "for example, to uphold home care services for residents on the island" (Umeå Kommun, 2024). Rather than prompting new organizational solutions adapted to Holmön's conditions, this combination of centralization and transport dependence instead becomes a reason why standard services are not provided.

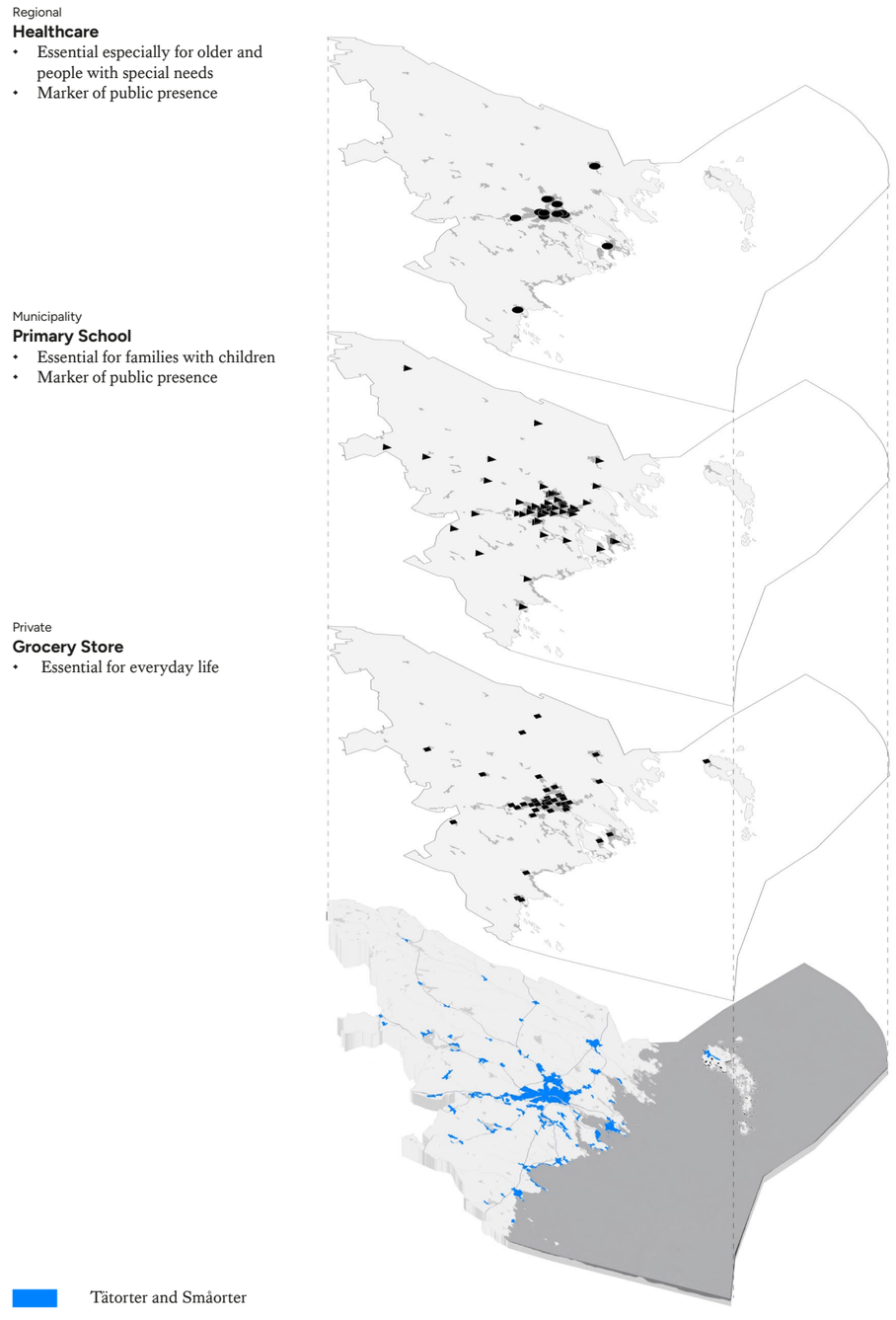
In contrast to this is Holmöns Lanthandel that points to a different logic. The grocery store is run by the community as an economic association, with some regional support, and can therefore operate at a scale and with routines that fit the island (Bygd och natur, 2024; Holmön.se). Unlike schools or healthcare, it is not bound by the same rigid frameworks and can be adjusted locally as circumstances change. The contrast between the flexibility of this community-run service and the inflexibility of public welfare structures illustrates how current systems struggle to accommodate places like Holmön, even when there is both local capacity and clear need.

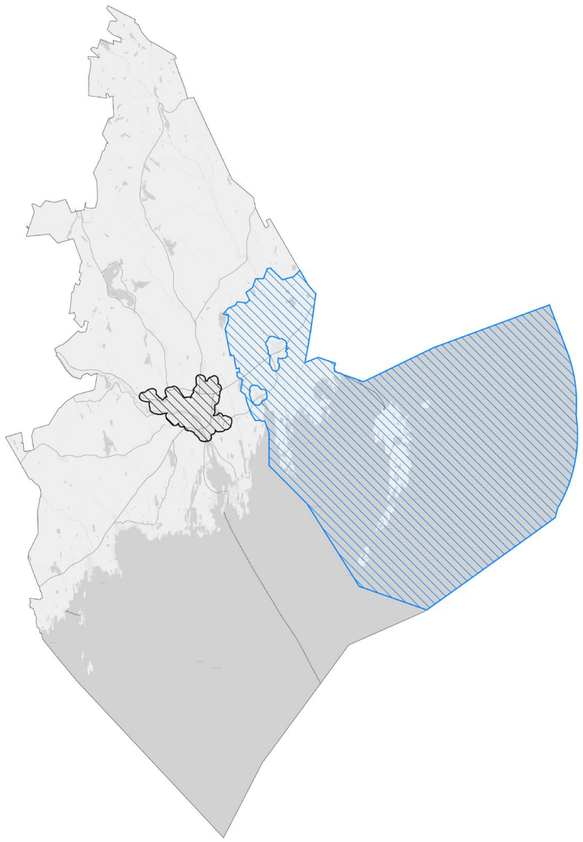
Transportation-dependent systems.

"The service that the public sector can currently offer residents of Holmön depends on functioning transport, which becomes a major vulnerability when existing infrastructure fails and no alternative solutions are available. This also applies to rural areas more broadly."

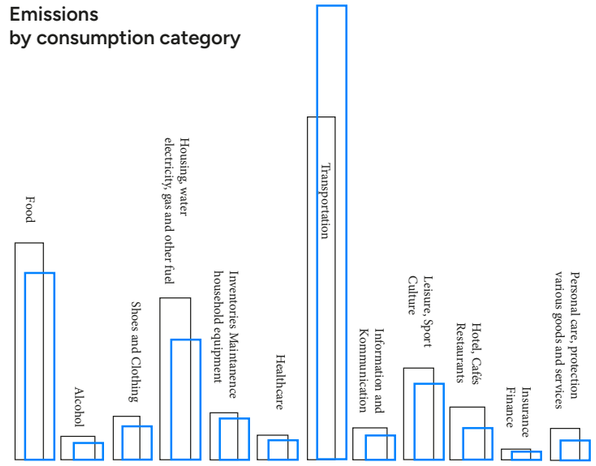
Holmön sits in an "urban" municipality on paper, but in practice it is a sparsely populated area that depends on transportation to reach almost all essential services. When infrastructure fails or regulations tighten, access to welfare functions becomes fragile, not only on Holmön but across rural Västerbotten. Transportation is therefore not just a practical question of getting from A to B. It is a structural dependency that shapes who can live here and under what conditions.

State-provided transportation to and from the island is organized through national regulations that leave little room for local adaptation, which in turn creates further isolation and uncertainty (Bergvall Kalén and Otterbjörk, 2025). Extreme weather and changing ice conditions make this even more vulnerable. When ice forms, the ferry stops; the hovercraft cannot operate until the ice is thick enough; helicopters are only brought in after several days of standstill and only on weekdays (Alexandersson and Bergvall Kalén, 2025). At the same time, key services such as school and healthcare have been centralized to larger towns. On the mainland, this can sometimes be compensated for by longer car journeys, but for Holmön the combination of centralization and an unreliable link means that the basic "life cycle" of services is broken.





This transportation dependency also appears in emissions data. Consumption-based statistics show that in Sävar's hinterland and Holmön, the transportation sector is the largest contributor to the local carbon footprint, while overall consumption-based emissions per capita are lower there than in central Umeå (Konsumtionskompassen, 2022). This suggests that everyday life in these rural areas is less carbon-intensive in terms of local consumption, but is weighed down by the need to travel long distances for basic services. Travel surveys for Holmön show that among full-time residents, the most common reason for commuting to and from the island is essential daily errands, such as medical appointments or administrative visits (Umeå kommun, 2023, p. 13). This also relates to the Swedish commuter tax deduction (reseavdrag) represents a substantial loss of tax revenue each year; earlier government estimates put the cost at around 4–5 billion SEK annually (Bergström, 2011). For the 2024 income year, commuters claimed travel deductions totaling just over 28.5 billion SEK, spread across 873,917 taxpayers (Swedish Tax Agency data reported in Carup, 2026)



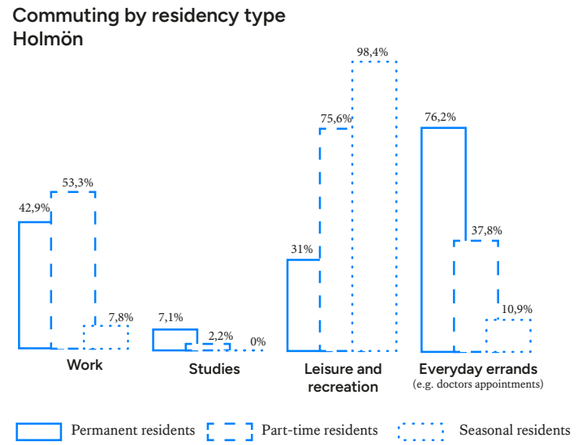
Total Consumption

Umeå: 5 549 CO_2 kg/person
Sävar Holmön: 5360 CO_2 kg/person

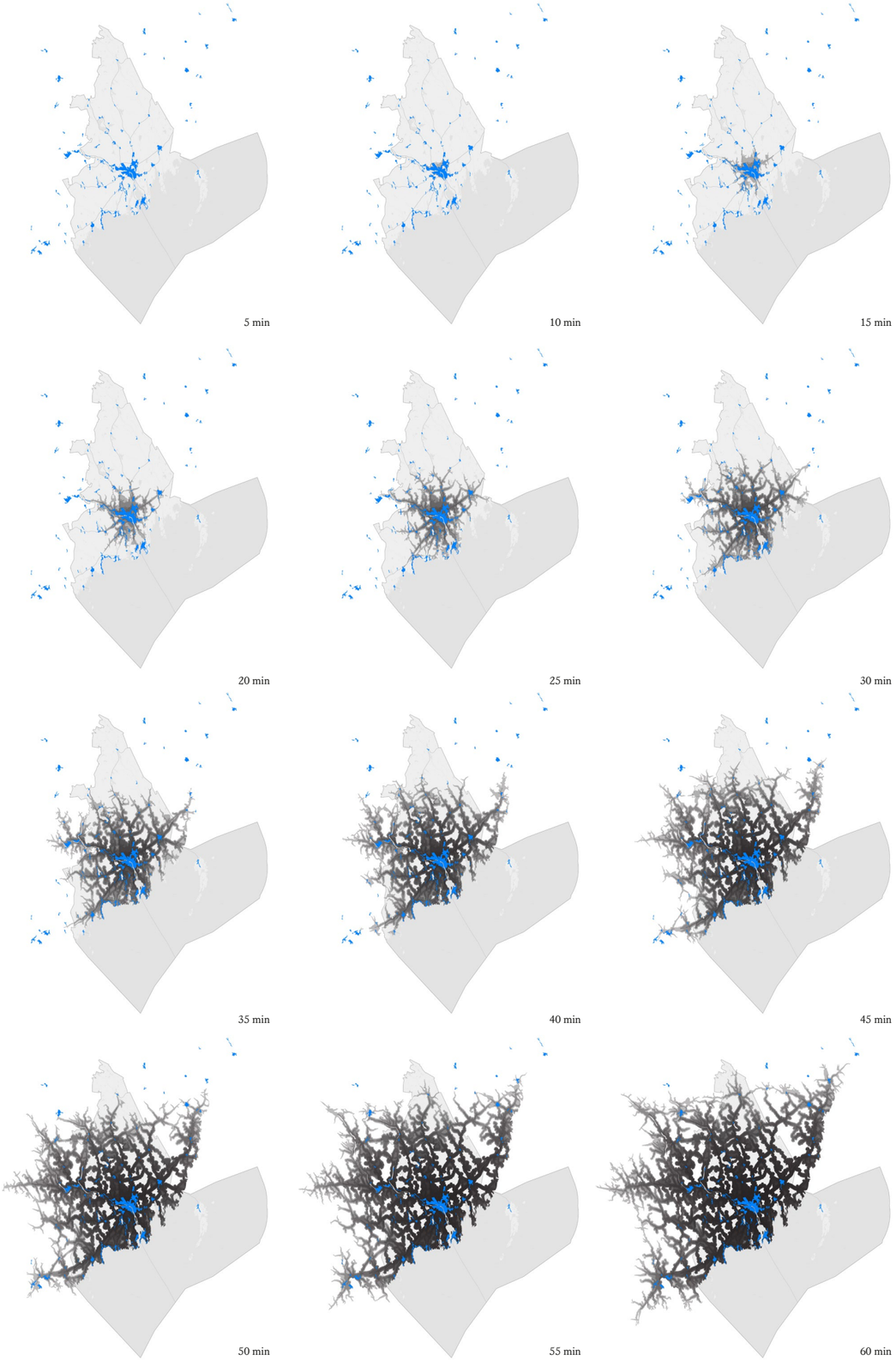
Without transportation

Umeå: 3 789 CO_2 kg/person
Sävar Holmön: 2965 CO_2 kg/person

Data: SEI (Swedish Environment Institute) 2023



Data: Umeå Municipality (Holmötrafiken – behov idag och imorgon) 2023



Testing of scalable service models

Testing scalable models: Med ön som katalysator steg *2*
Building on the preliminary study Med ön som katalysator, a second phase has been launched under the title Med ön som katalysator steg 2, described as a testbed for citizen-driven welfare models and also run in collaboration between Umeå Municipality, Region Västerbotten and Holmöns Utvecklingsforum. The project is funded through the national Impact Innovation program, a joint initiative by the Swedish Energy Agency, Formas and Vinnova based on Agenda 2030 and national mission-oriented innovation goals (Impact Innovation, 2024). On Holmön, the project tests concrete arrangements for citizen-driven welfare, where key components include a samhällsrum (community room) at Prästgården, safety alarms (trygghetslarm), a local coordinator position, and the voluntary role “bybo i beredskap”.

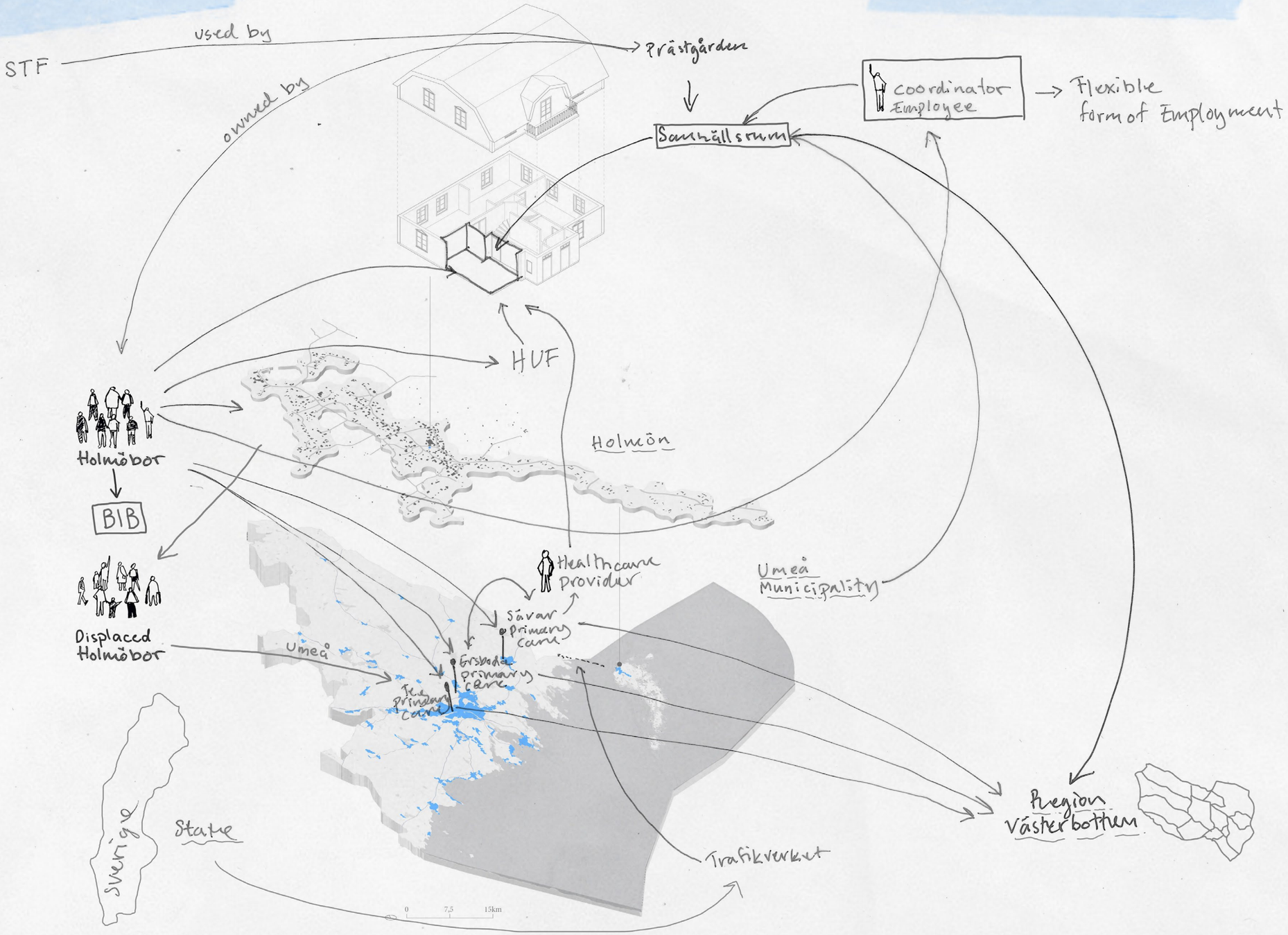
The samhällsrum builds on earlier “hälsorum” initiatives in the region, but expands the program from healthcare alone to a broader combination of digital care access, local support, and civil preparedness coordinated in one place (Bergvall Kalén and Otterbjörk, 2026). It is to be implemented on Holmön in Prästgården, a former vicarage that the local community acquired collectively through crowdfunding. Prästgården is currently driven by Holmöns bygdebolag AB used for STF overnight stays. Positioning the samhällsrum in this building connects new welfare functions to an already active community resource and tests how an existing rural structure can be reprogrammed as shared infrastructure for services.

Through this ongoing project, and particularly through meetings with the project group and a community workshop at Prästgården, Holmön has also become my entry point into the wider structural conditions discussed in this thesis.

Workshop insights

Within Med ön som katalysator steg 2, I participated in a workshop at Prästgården in 2026 with mainly older residents, where the municipality and project group gathered input on how the planned samhällsrum and coordinator role could support everyday life on Holmön (author’s field notes, 2026; Umeå kommun, 2025). The workshop combined a short questionnaire with group discussions about which functions and services different groups on the island would need, how the room might be used beyond healthcare, and what additional activities and social programs should be hosted at Prästgården.

The discussions made the service gap concrete. Half of the ten participants stated that under current conditions, they could not live on Holmön year-round and instead stayed mainly during the summer months (author’s field notes, 2026). One participant described how her orthopedic surgeon would not operate as long as she was registered on Holmön because rehabilitation on the island was considered impossible (author’s field notes, 2026). These accounts underline that the planned samhällsrum is for some a matter of if it is possible for them to live there or not. It becomes an attempt to rebuild basic infrastructures of care and support on the island and, by extension, to restore some degree of everyday autonomy.


STF
used by
owned by
Prästgården
Samhällsrum
coordinator
Employee
→ Flexible
form of Employment
HUF
Holmön
Holmöbor
BIB
Displaced
Holmöbor
Healthcare
providor
Umeå
Municipality
Sävar
primary
care
Umeå
Ersboda
primary
care
Tegs
primary
care
Sverige
State
Trafikverket
Region
Västerbotten
0
7,5
15km

Part 3

Rural livability as a shared interest



Why, then, should people still be able to live in rural areas? Many leave the countryside in search of work, education and, in some respects, better living conditions. According to United Nations projections, around 68 percent of the world's population will live in urban areas by 2050. Wouldn't it, at first glance, seem more efficient if everyone simply moved to the city?

Both city and countryside are necessary components of a sustainable society – not as opposites, but as interdependent territories whose futures are tightly intertwined. The Swedish Board of Agriculture's report Allt om miljö och hållbarhet på landsbygden emphasizes that a large share of the resources sustaining urban life – including food, biomass, energy and key ecosystem services – are produced and managed in rural areas, and that a living countryside is therefore a precondition for a sustainable city (Jordbruksverket, 2013).

"Freedom of choice"

It is often assumed that the concentration of the population in cities directly reflects people's preferences and, by extension, justifies the way services are distributed. Yet while services tend to follow population, survey data on residential preferences tell a different story. Data from the last two years suggest a disconnect between where people live and where they would actually like to be. In a 2024 survey commissioned by Fastighetsbyrån, the countryside was ranked as the place people would most like to live the next time they move, ahead of both mid-sized towns and the largest cities. More recent findings from the 2025 survey show a slight shift, with big cities at 21 percent and the countryside at 20 percent; the two remain effectively tied as the most preferred options (Fastighetsbyrån, 2024; Fastighetsbyrån, 2025).

Regardless of which of these years we're looking at, the figures do not reflect the actual distribution of the Swedish population, where over 70% of the population lives in tätorter with more than 10 000 inhabitants (SCB, 2026). This gap reveals a systemic disconnect. The results further show that safety and security have become the most important criteria for choosing a location, with Johan Engström, CEO of Fastighetsbyrån, noting that growing societal insecurity is fundamentally altering housing preferences (Fastighetsbyrån 2024).

Furthermore, the demographic trend of moving to the countryside is rarely a move into the unknown. Data indicate that those moving to rural areas are predominantly individuals who grew up there, returning to their roots (SVT Nyheter 2024). This "return-migration" highlights a critical point for regional development: the rural population is not a static group, but a mobile one that seeks to return when life stages permit. A large-scale study from Umeå University focusing on Northern Sweden (2025) confirms that this is a significant regional phenomenon; for these individuals, building a family, community, and closeness to nature is especially important to them in their desire to go back to their hometown village in the countryside (Ståhl, 2025).

The fact that many small towns and rural areas, continue to shrink despite these preferences points back to the structural barriers identified on Holmön. If a family desires the safety and space of the countryside but cannot find a school or a reliable commute, living in rural areas becomes functionally impossible. In this context, the "freedom to choose" is a hollow concept unless the physical and service infrastructure is redistributed to support it. Living in rural areas has to some extent become less a matter of personal choice and more a consequence of systematic neglect: why live here when it is made so difficult?

Urban strains

The logic of centralization can be defended through the lens of efficiency. The traditional metropolitan model is often considered the most sustainable option due to its higher residential density, shared infrastructure, and shorter travel distances. While this is true, the city also isn't perfect in how it operates today. Facing issues of ecological, social, and infrastructural nature.

The pursuit of urban growth leads either to the reevaluation of green spaces in our cities or to urban sprawl. Urban sprawl occurs if growth is not contained and spills into the hinterland. This increases car dependency, fragments rural landscapes, and can be at the cost of nature or productive farmland (Kraft, 2023). The other alternative avoids sprawl by densifying, which usually means building on every available plot and often sacrificing the inner-city nature that provides cooling, water management, and well-being for residents.

Umeå toward 200 000
This is clearly visible in the Umeå metropolitan area, where the municipality has set a strategic goal to reach a population of 200,000 by 2050 (Umeå Kommun, 2024). The municipality states that this growth goal is necessary to maintain a high level of service, a diverse labor market, and a robust tax base. In their reasoning, Umeå Municipality highlights many of the same systemic problems brought up in this thesis, such as the need to sustain the welfare state and ensure a high quality of life. However, by focusing so heavily on the urban core as the sole engine to solve these

problems, the strategy fails to see the full territorial picture.

To reach this goal, many new housing areas are planned for sites that currently function as key green areas for the city. Developments like Tomtebo Strand and Norra Ön are both examples of this. The development of Norra Ön, in particular, has been challenged by the County Administrative Board (Länsstyrelsen) because the central island is an important natural environment and habitat for the white-backed woodpecker (vitryggig hackspett) (Umeå Kommun, 2026).

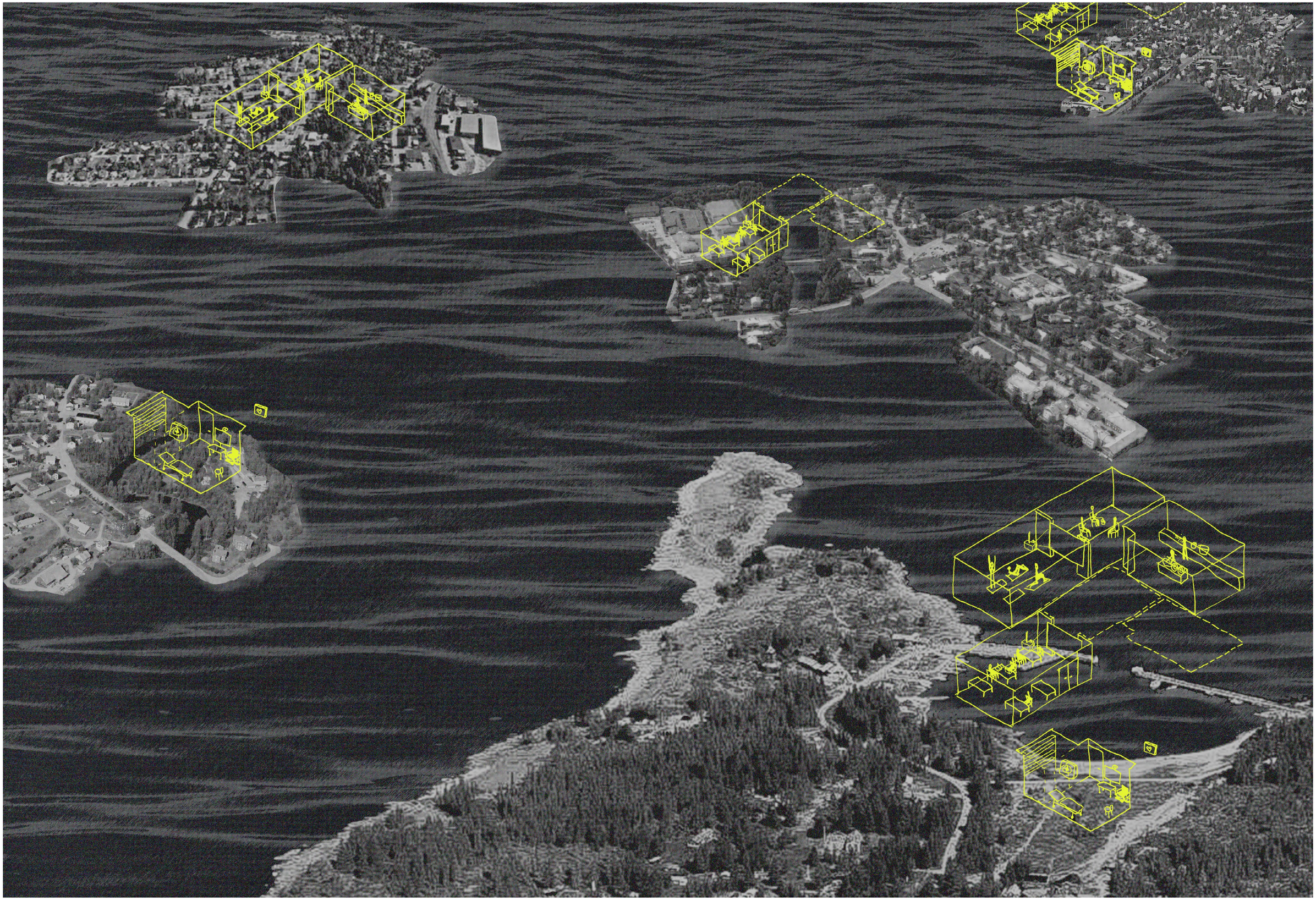
Umeå municipality's growth scenario suggests that this expansion is not just about the city but is a strategy for the entire region. The municipality argues that Northern Sweden needs a "regional motor" and a strong urban alternative to meet ongoing urbanization (Umeå Kommun 2024). This implies that urban growth in Umeå municipality, an already dominant force in the region, will benefit the entire region and assumes that a bigger city and stronger storstadskommun creates a stronger region. However, as seen in the case of Holmön and its absorption into umeå municipality, centralization of services and transportation dependencies, this might not be a foolproof strategy. This thesis questions the fundamental assumption of this strategy: Does the region actually need more urban growth to become more resilient, or are there other ways to reach that goal?

Umeå Municipality's strategy of reaching 200 000 also assumes that we need to build our way to growth. In many major cities, there is a housing shortage, including Umeå, which has reported a housing shortage for a long time, especially affecting vulnerable groups such as students, the elderly, people with disabilities, and immigrants (Länsstyrelsen Västerbotten, 2025, p.26-30). This scarcity is used to justify building new and building away green spaces in Umeå. When we look at the broader regional map, the story changes completely. Recent data highlights a significant number of vacant apartments across various Swedish municipalities (Thambert, Nordström and Westholm 2026). While Umeå struggles to house its students, nearby rural municipalities are burdened by the costs of maintaining or even demolishing empty housing stock. For these smaller towns, vacant buildings are not just "unused space" but a direct financial strain on their municipal or business budgets.

From an environmental perspective, the most sustainable building is the one that already exists. Construction is one of the most carbon-intensive industries in the world (UN Environment Programme, 2025). Every new apartment block in Tomtebo Strand represents a massive expenditure of energy and raw materials, while existing homes in the rural hinterland sit empty. In the context of Umeå municipality's positioning as the 200 000 goal as a regional strategy, this approach warrants critical examination: why is building new the first response? Prioritizing the reuse and renovation of existing housing stock is not just a common-sense economic strategy; it is a fundamental requirement for climate resilience.

Community and Resilience

Research shows that strong communities and social capital enable faster recovery during social disturbances. Rural areas generally have stronger communities, and people tend to know their neighbors better, understand each other's strengths and weaknesses, and know who has access to which tools or skills, making cooperation easier (Laestadius 2024). This is also connected to the limited public presence in rural areas, which fosters a mindset of self-reliance and mutual support. Veronica Strandh (2021), a researcher at Umeå University, emphasizes that community engagement is a key piece in crisis preparedness; for example, knowing your neighbors facilitates collaboration and mutual aid during disruptions. This suggests that the rural is a form of everyday organization that can become especially valuable in times of disruption.



Part 4

“Urban” Planning Discourse and Redistribution

Redistribution by Urban design

Adrià Carbonell’s dissertation, Redistribution by Urban Design: Centrality, Ecology, and Transition in Järvafältet, 1930–2030, frames redistribution as a territorial response to the limits of the traditional metropolitan model and as a disciplinary shift away from growth. Instead of continuing to concentrate value, services, networks, and infrastructures in already advantaged cores, planning should redistribute existing resources across the territory (Carbonell, 2025, p. 393). This shifts attention from the compact city alone to the wider urban territory, where geography, ecology, land use, and political control over land come together (Carbonell, 2025, p. 393).

In this sense, the city is not separate from its surroundings, but part of a larger territorial unit in which urban and rural conditions are intertwined. Carbonell describes the urban crisis as two-sided. On one hand, the traditional metropolitan model produces internal strains such as inaccessible housing, long commutes, and high energy and carbon footprints (Carbonell, 2025, p. 395). On the other hand, it pushes costs and pressures outward, where non-city territories are increasingly instrumentalised as economic, ecological, and logistical resources to sustain urban life (Carbonell, 2025, p. 372). In this sense, the city’s problems and the hinterland’s burdens are part of the same territorial system.

For this thesis, Carbonell’s dissertation is relevant because it reveals the relation between city and countryside and makes redistribution a question of how resources and burdens can be shared more equitably across the whole territory, rather than being concentrated in one part of it (Carbonell, 2025, pp. 393–395).

A Pattern Language

An earlier contribution to planning discourse that engages with questions of distribution in a more spatial form is Christopher Alexander’s A Pattern Language (1977).

Pattern 2, The Distribution of Towns, can be understood as an attempt to describe a balanced regional structure by giving it a spatial form, through metrics such as population size and how settlements of different sizes are distributed in relation to each other across the territory.If a region is dominated by small villages, it cannot support the functions associated with modern society. If it is dominated by large cities, the region begins to break down because people are no longer located where they are needed to take care of it. The aim is therefore a distribution of settlements of different sizes across the territory. A few larger cities are supported by a greater number of towns and an even larger number of small settlements, all spread relatively evenly across the land. This is not something that happens on its own, but needs to be encouraged through regional initiatives such as zoning, land allocation, and incentives that influence where services and industries locate (Alexander, Ishikawa and Silverstein, 1977, p. 16-20).

This connects directly to Pattern 6, Country Towns, which focuses on the difficulty of maintaining smaller settlements in relation to larger urban centers. The city acts as a strong attractor, often pulling resources and people away from smaller towns. Therefore the pattern calls for for preserving existing towns and supporting the development of new ones that are self-contained. These towns should have enough population and economic base to support everyday life locally, rather than functioning primarily as residential extensions of a larger city (Alexander, Ishikawa and Silverstein, 1977, p. 33-36).

In the context of Västerbotten, the exact numbers proposed by Alexander, such as population sizes or distances between settlements, are not directly applicable. The geography is too sparse and the settlement structure too uneven. However, the underlying idea of working toward a more balanced distribution remains relevant. The patterns describe a way of thinking about rural areas as autonomous parts of a larger system, rather than as dependent peripheries.

Pattern 26, Life Cycle, adds another dimension to this by focusing on the internal composition of communities. Alexander argues that each settlement should include a balance of people across different stages of life, along with the spaces and institutions that support them. A community that lacks this balance becomes incomplete and increasingly dependent on others (Alexander, Ishikawa and Silverstein, 1977, p. 139-145).

This becomes particularly important in relation to service distribution. In Umeå Municipality, redistributing services cannot simply mean extending urban functions outward. If rural settlements are to remain viable, they need to support a broader range of everyday functions locally.

Speculative design Method

Speculative design is framed by the idea that things could be radically different, and that imagining such difference can help make alternative worlds feel possible. Rather than predicting the future, it aims to provoke possible futures and use them as tools to discuss what people actually want the future to be like. It usually takes the form of scenarios, often beginning with the question “What if,” and deliberately moves away from what is merely possible. Because speculative design treats imagined scenarios as instruments for debate, it deliberately sets aside present-day constraints and allows us to explore futures that would otherwise be dismissed as impractical (Dunne and Raby, 2013). This also connects to the idea of Utopia “We need utopian thinking because without it, we are constrained by the tyranny of the possible” (Duncombe, Peters, 2012).

Part 5

Speculative Redistribution Strategy

Islands on Land

While the binary definition of rural in Umeå Municipality enables the targeted distribution of limited resources, it also reinforces a fixed urban-rural logic that fails to capture the nuances of the landscape or the distinct challenges faced by individual settlements. Islands on Land is a new territorial definition in which all settlements in Umeå Municipality are understood as islands on land, regardless of population size or geographic location. Their boundaries are drawn from SCB's definitions of tätorter and småorter, creating a framework that reframes the municipality as an archipelago of semi-autonomous islands. This gives Holmön, as a geographic island, a clearer spatial and conceptual agency as a pilot case. Shifting the planning focus away from "fixing" rurality in relation to the urban system and towards strengthening of local autonomy.

This autonomy is not exclusive to rural areas; it can also benefit urban islands on land by encouraging them to function more independently within a larger network. In this framework, the main measure of success is not proximity to a center, but the local capacity to support jobs, basic needs, leisure, community life, and essential services.

Speculative Redistribution Strategy

The speculative redistribution strategy explores how welfare services could be redistributed across Umeå Municipality within the Islands on Land framework. It works specifically with primary schools and healthcare nodes, since these are essential services and visible markers of public presence in the territory. Rather than proposing a final solution, it serves as a design tool to question the current spatial logic of service provision and to test what kinds of territorial conditions might emerge if access were redistributed according to other criteria. The strategy is developed in four stages, moving from identifying existing gaps to translating redistributed access into architectural and programmatic form. In this sense, the numbered sections describe both the strategy itself and the process through which it was developed.

Redistribution Strategy

1. Identify current distribution gaps
Map the existing distribution of essential services

2. Apply the redistribution tool
Use the code-based tool to redistribute the service based on new criteria of population and distance.

3. Locate residual service gaps
Use the service buffer zones to identify settlements that still fall outside adequate access. These underserved islands become the places where additional service points can be added.

4. Translate redistribution into spatial implementation
Determine what architectural scale is required for each intervention and identify buildings that can host redistributed services through readaptation.

Identify current distribution gaps

To understand how welfare services are currently distributed across Umeå Municipality, I mapped primary schools, healthcare facilities, and grocery stores using information from skolkoll and region västerbotten (KÄLLA). These services were selected because they function as essential markers of public presence and form part of the everyday infrastructure that makes settlements viable. The resulting maps and axonometric diagrams show that services are not evenly distributed across the municipality, but are concentrated in the larger tätorter, especially Umeå, Holmsund and Sävar. Peripheral settlements such as Holmön remain underserved, revealing a spatial hierarchy in which access to basic services follows centrality rather than territorial need.

Applying the redistribution tool

The redistribution tool is a speculative test of how welfare services might be placed across the Islands on Land framework according to a different spatial logic. Developed as a QGIS Python script through prompting with Claude AI, the tool uses two inputs: population and distance to centrality. Population captures where demand is greatest, while distance captures geographical disadvantage by giving weight to settlements that are further away from the municipality's central nodes. Together, these inputs create a more equitable basis for redistribution than population alone, since they begin to balance need with territorial disadvantage.

During the iterative development of the script, it became clear that a simple normalization of the two values produced overly polarized results, with too much weight drawn towards Umeå and too little distributed across the middle territories. To correct this, the tool was adjusted by introducing a maximum number of points per population band and by adding buffer zones to prevent nodes from clustering too closely together. Testing different buffer distances showed that 15 km produced workable healthcare islands, while 5 km was more appropriate for primary school coverage. In this way, the buffer zones do not only regulate spacing; they also define territorial service islands and translate the logic of the 15-minute city into a scaled territorial condition suited to the municipality's dispersed geography.

To see the full code for the redistribution tool, see Appendix 1

Redistribution Tool

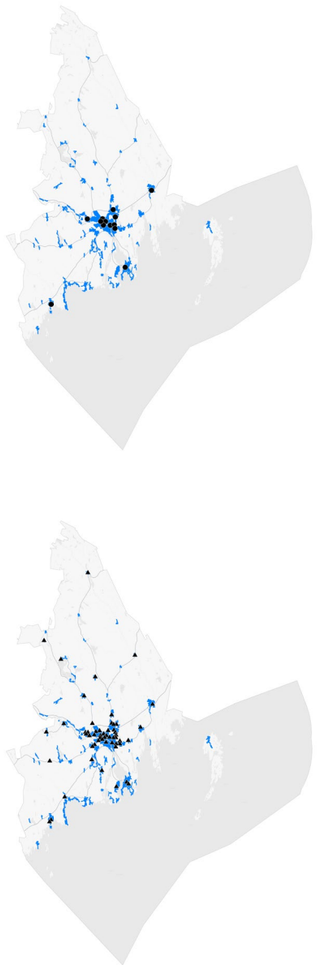


13 healthcare facilities, including hospitals and clinics, regional and private

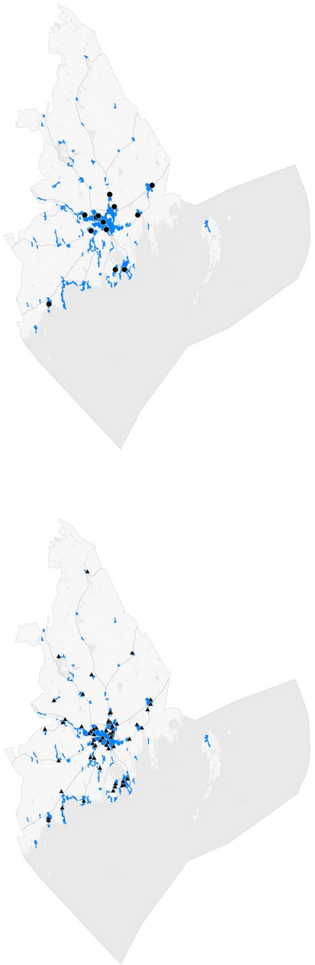


64 primary schools, including private and municipal providers

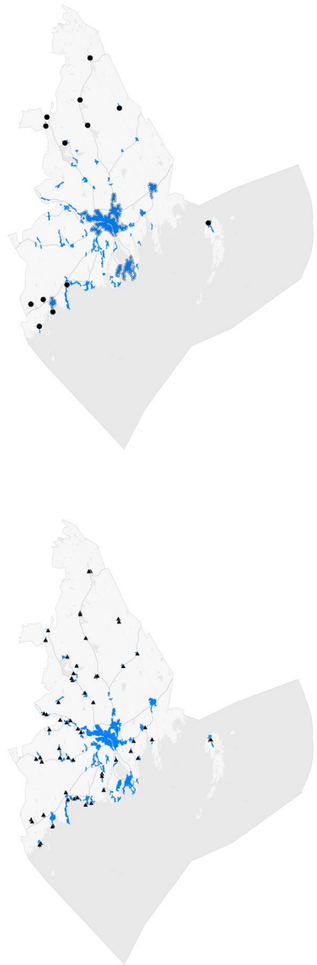
Current



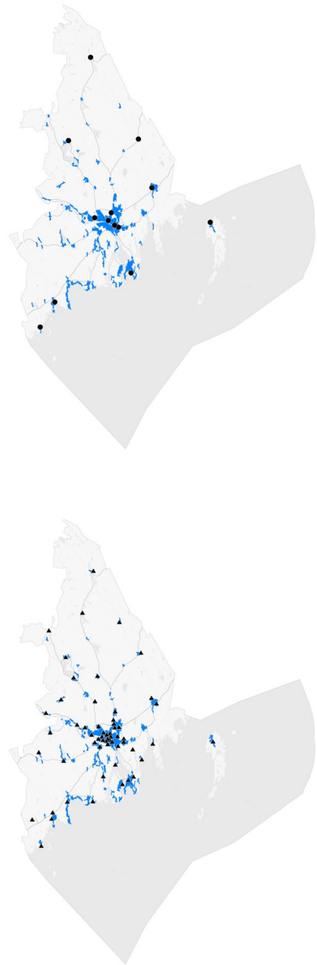
Population



Distance to Centrality



Distance 50%
Population 50%



Locating residual service gaps

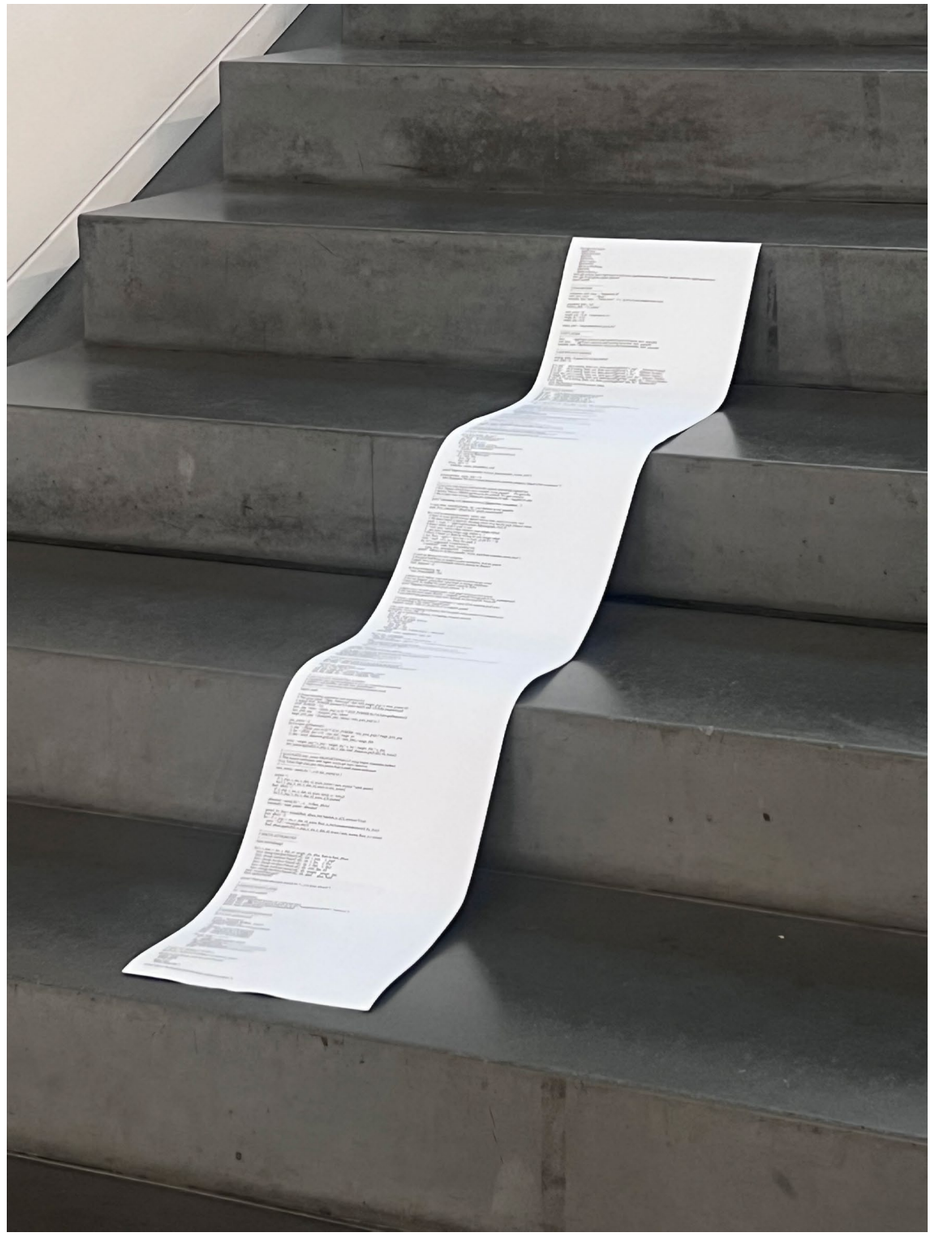
After the redistribution tool has been applied, the resulting service buffers are used to locate the settlements that still fall outside adequate access. These residual gaps reveal which islands remain underserved even after redistribution and therefore become the places where additional service points may be needed. The point is not simply to distribute services evenly, but to show where the remaining spatial inequalities are concentrated and where a second layer of intervention is required. In this way, the buffer zones help identify the limits of the first redistribution and make the next step of the strategy possible.

Translating Redistribution into Spatial Form

The output map should be understood as schematic rather than literal: all redistributed service nodes are marked with the same-sized points, even though the services they represent would operate at very different scales in reality. A point on the map may therefore stand for anything from a major regional facility to a much smaller local service node. For that reason, the map is not a final design but a way of identifying where service capacity might be placed and how different territorial conditions could be supported. The Samhällsrum is used here as the smallest possible scale of a service node, but the program is not taken over as a fixed model. Instead, the initial idea of the Samhällsrum in the regional project, together with the workshop perspective that it should support additional functions, becomes the basis for a further developed program. In other words, the project evolves the Samhällsrum from a healthcare room into a broader spatial typology that can also accommodate other functions.

Samhällsrum as Spatial Prototype
The Samhällsrum is first and foremost a response to the lack of healthcare on the island: a space for digital appointments, first aid, emergency equipment and occasional visits from medical staff. In this way, one professional can travel to the island instead of many residents travelling away from it, while also contributing to the local economy. The coordinator does not function as a healthcare worker, but as someone who supports residents by opening, maintaining and assisting with the space. The room does not replace healthcare, but gives it a local foothold again. At the same time, the workshop made clear that the Samhällsrum should not be limited to health alone; both the community and the project group agreed that it should support additional functions, since on an island everything is connected.

A second function addresses schooling and begins to support a fuller life cycle on the island. In response to workshop input that healthcare and other functions should remain separate for reasons of emergency access, privacy and integrity, the school function is expanded into the current matsal in the prästgård. This means that the Samhällsrum is no longer only a single room, but begins to take on a larger physical form where the educational program can occupy a separate space. Children may still attend school in Sävar, but when the ferry is canceled, this space can serve as a temporary substitute. It is not a complete school solution, but it lowers the threshold for families to remain on the island and could, over time, become a stepping stone toward a permanent school — or not.

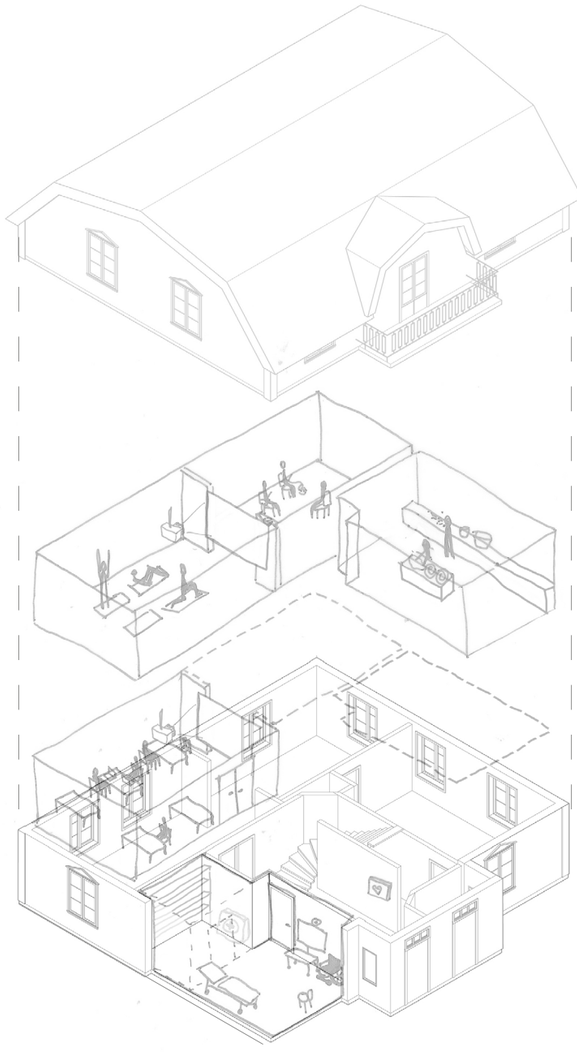


Early iteration of the code for the redistribution tool, developed with Claude AI. Printed version.

The third form extends beyond service access and into the public life that the municipality and region are also responsible for supporting through regional development, culture and leisure. It is also separate from the healthcare functions, but overlaps with the school functions. It hosts civic functions and activities that the community already partly organises itself, providing a designated place for them to gather and take place. In this way, the Samhällsrum is activated not only during the day, but also in the evenings and at weekends. This multi-functional approach is also practical: a healthcare room alone cannot sustain a full-time coordinator for a population of 60 people. By hosting several programs under one roof, the role becomes viable, the space remains active, and the samhällsrum gives physical form to the collaboration between the local community, the municipality, and the region.

The Samhällsrum is not an isolated object, but part of the redistribution strategy. With Prästgården as its first implementation, it combines health, school and community functions in response to the spatial demands of service redistribution. The strategy is not limited to one building, but can be adapted across other islands by reactivating underused structures with similar scales and characters. Former centrum buildings, bygdegårdar, vacant housing and church-owned premises all represent existing spatial capacity that can be readapted rather than replaced. In this sense, the project identifies typologies for reuse that already have a social or civic role in the landscape, but that can be reprogrammed to host redistributed services.

Samhällsrum



Adaptive reuse



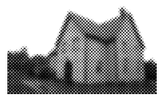
Centre Facilities | Botsmark
Previously Handlarn grocery store.
currently emtpy



Vacant housing | Tavelsjö
Empty apartment for rent.



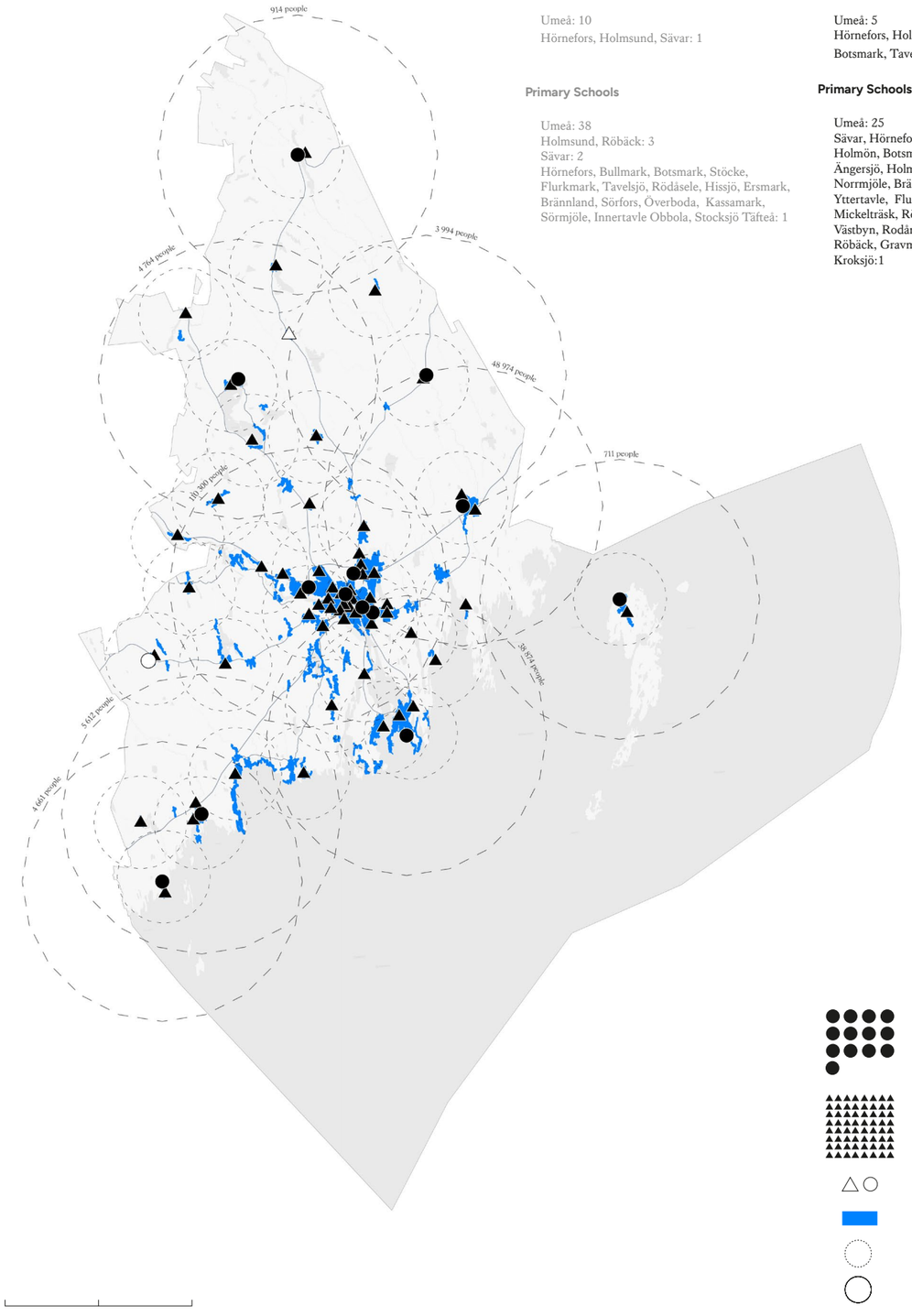
Community infrastructure | Holmön
Prästgården owned and driven by the community



Church-owned premises | Hössjö
Hössjö missionshus religious and community building used for worship, gatherings, and seasonal social functions.



Community infrastructure | Norrbyn
The old school building currently used as the community meetingplace, *Träffpunkten.*



Current Distribution

Healthcare

Umeå: 10
Hörnefors, Holmsund, Sävar: 1

Primary Schools

Umeå: 38
Holmsund, Röbäck: 3
Sävar: 2
Hörnefors, Bullmark, Botsmark, Stöcke, Flurkmark, Tavelsjö, Rödåsele, Hissjö, Ersmark, Brännland, Sörfors, Överboda, Kassamark, Sörmjöle, Innertavle Obbola, Stocksjö Täfteå: 1

Redistribution 50/50

Healthcare

Umeå: 5
Hörnefors, Holmsund, Sävar, Norrbyn, Bullmark, Botsmark, Tavelsjö, Holmön Hössjö.: 1

Primary Schools

Umeå: 25
Sävar, Hörnefors, Holmsund, Röbäck: 2
Holmön, Botsmark, Norrbyn, Bullmark, Tavelsjö, Ängersjö, Holmnäs, Yttersjö, Överboda, Ström, Norrmjöle, Brännäset och Boviken, Sillviken, Yttertavle, Flurkmark, Kassjö, Haddingen, Mickelträsk, Rödåsele, Rovögen, Gubböle, Västbyn, Rodånäs, Brännland, Sand, Ersmark, Röbäck, Gravmark, Ansmark, Fäbodåkern, Kroksjö:1

Essential Services
13 healthcare providers including hospitals and clinics private and regional providers

64 primary schools, including private and municipal providers

Identified as underserved

Island on Land

5km buffer zone

15km buffer zone

Part 6

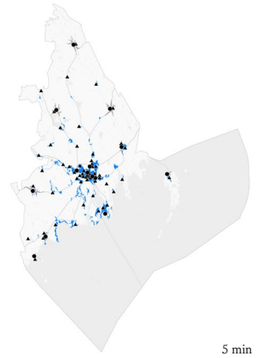
After Redistribution

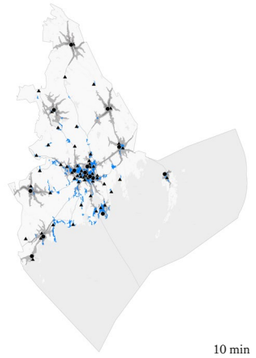
After redistribution, a different spatial order begins to appear. The new distribution shows a new logic of equality, where the service buffers no longer cluster around the urban core but begin to overlap across the municipality. This does not produce a fully new territory, but it reveals that access can be organized through a different spatial logic than the one currently in place. The buffer zones also define the scale of the redistributed service islands. Testing different distances showed that 15 km produced workable healthcare islands, while 5 km was more appropriate for primary school coverage. Rather than functioning as abstract circles on a map, these radii establish different territorial conditions for access, and their overlap reveals where redistributed services begin to form a more continuous spatial network.

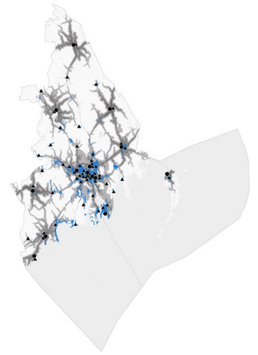
The change also affects the transportation network. As services are redistributed, travel no longer radiates only from Umeå, but begins to form a shorter and more distributed system of movement and access. In the current system, dependence on transportation is clear. Travel time in Umeå Municipality is shaped by the concentration of essential services in and around Umeå, where access is distributed less by territorial need than by proximity to the urban center. On land, if Umeå is treated as the central service provider, the calculated travel time to reach all other settlements is 55 minutes. Even within this calculation, Holmön is excluded, meaning that reaching it requires additional and highly unpredictable travel time.

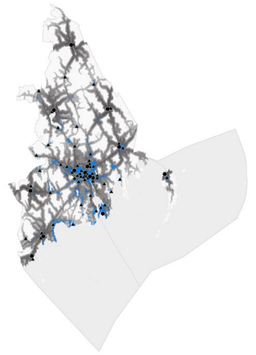
The redistribution strategy changes this condition. By moving services across the territory, the network is shortened from 55 to 20 minutes. The roads have not changed, but the service pattern has, and the carbon penalty of distance begins to shrink. In this way, redistribution is not only a social project, but an environmental one as well.

For Holmön, this is the first visible step in a broader ripple effect, where redistributed services support the return of displaced Holmöbor, reactivate everyday life, and gradually strengthen local economy and autonomy.


5 min


10 min


15 min


20 min

Discussion

The redistribution strategy presented in this thesis does not solve the structural conditions that produce rural decline, but it makes those conditions spatially legible and therefore open to critique. By shifting the location of essential services, the project challenges the assumption that access should concentrate around urban centrality and instead proposes a more distributed territorial logic. In this sense, the project is not only about Holmön, but about how planning could begin to imagine welfare as something that is shared across territory rather than pulled toward a single center.

At the same time, redistribution of services alone is not enough to create true territorial autonomy. A functioning community requires more than access to schools and healthcare; it also needs jobs, governance, everyday support systems, and a stable relation between local capacity and public responsibility. This is where the project becomes a reminder that spatial intervention cannot stand outside the broader ecosystem it tries to influence. Services become a cornerstone, but they cannot carry the entire structure by themselves.

The project also reveals its own limits. Umeå Municipality does not exist as an island in itself, but is directly connected to neighboring municipalities with their own islands on land, where these systemic challenges are often even more pronounced. This interdependence raises the question of how redistribution could operate across administrative borders rather than being confined by artificial lines. At a regional scale, where geographic distances grow much larger, the need for a collaborative territorial approach becomes even more urgent. Expanding the strategy beyond municipal boundaries would allow redistribution to reach further, transforming it from a municipal tool into the basis for a more resilient regional network.

The strategy also opens up a wider discussion about sustainability, where local livability becomes the starting point for rethinking regional development. Systematically providing services to remote and rural areas does more than reduce forced transportation; it helps these communities remain viable. This shifts the perspective on regional growth: instead of assuming that city expansion and new housing construction are always the answer, as in Umeå's growth plans, the remote-settlement lens asks why new infrastructure should be built when substantial vacant housing stock and existing structures already remain underused throughout the wider region (Thambert, Nordström and Westholm, 2026).

Securing the long-term viability of these peripheral places opens up a broader understanding of sustainability. When rural communities are supported rather than left to decline, future life can be shaped by a combination of digital, place-independent work and local practices grounded in repair, cultivation, and regional resources. Living rurally means maintaining a direct connection to land, weather, and seasons. Spatial redistribution preserves this ecological perspective by preserving those perspectives and shifting decision-making power from the city to the Islands.

To conclude I'd like to end with a question that has been like a mantra throughout the process of making this thesis:

What is the role of the architect?




Apotek
HOLMÖNS LANTHANDEL
Café



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Visual Material:
All Visual Material is produced by the author

Appendix

Qgis python code for “Redistribution Tool”

```
from qgis.core import (
    QgsProject,
    QgsVectorLayer,
    QgsField,
    QgsFeature,
    QgsGeometry,
    QgsPointXY,
    QgsSpatialIndex,
)
from qgis.analysis import (
    QgsVectorLayerDirector,
    QgsNetworkDistanceStrategy,
    QgsGraphBuilder,
    QgsGraphAnalyzer,
)
from qgis.PyQt.QtCore import QVariant
# ------------------------------
# PARAMETERS  — edit these
# ------------------------------
settlements_layer_name = “Settlements all”
road_layer_name        = “Roads”
centrality_layer_name  = “Urban zones”

# Tolerance (metres) used to snap settlement centroids onto the
road network.
# Increase if settlements still show dist=0 or inf after a run.
SNAP_TOLERANCE = 10.0

# Settlements listed here are treated as maximally remote
regardless of what
# the road network returns. Use this for islands or ferry-only
settlements
# where the graph connects them via a ferry edge that should
not count as road.
ISLAND_SETTLEMENTS = [“Holmön”]
# ------------------------------
# GET LAYERS
# ------------------------------
layer          = QgsProject.instance().mapLayersByName(settle-
ments_layer_name)[0]
road_layer     = QgsProject.instance().mapLayersBy-
Name(road_layer_name)[0]
centrality_layer = QgsProject.instance().mapLayersBy-
Name(centrality_layer_name)[0]
# ------------------------------
# ENSURE road_dist FIELD EXISTS
# Always overwrite — add field only if missing, then write fresh
values below
# ------------------------------
if “road_dist” not in [f.name() for f in layer.fields()]:
    layer.dataProvider().addAttributes([QgsField(“road_dist”,
QVariant.Double)])
    layer.updateFields()

idx_road_dist = layer.fields().indexOf(“road_dist”)
if idx_road_dist == -1:
    raise Exception(“Field ‘road_dist’ not found after addAt-
tributes.”)
# ------------------------------
# IDENTIFY WHICH SETTLEMENTS OVERLAP A
CENTRALITY POLYGON
# Those get distance = 0 without needing network routing
# ------------------------------
print(“Checking which settlements overlap centrality poly-
gons...”)

centrality_geoms = [f.geometry() for f in centrality_layer.
getFeatures()]
features_list    = list(layer.getFeatures())

centrality_index = QgsSpatialIndex(centrality_layer.getFea-
tures())

overlapping_ids = set()
for f in features_list:
    candidates = centrality_index.intersects(f.geometry().
boundingBox())
    for cid in candidates:
        if f.geometry().intersects(centrality_geoms[cid]):
            overlapping_ids.add(f.id())
            break

non_overlapping = [f for f in features_list if f.id() not in over-
lapping_ids]

print(f”  {len(overlapping_ids)} settlements inside/overlapping a
centrality  dist = 0”)
print(f”  {len(non_overlapping)} settlements need road routing”)
# ------------------------------
# BUILD ROAD NETWORK GRAPH
# Only non-overlapping settlement centroids are tied in as extra
vertices;
# centrality coverage comes from native road vertices inside
each polygon.
# ------------------------------
print(“Building road network graph...”)

settlement_centroids = [f.geometry().centroid().asPoint() for f
in non_overlapping]

director = QgsVectorLayerDirector(
    road_layer, -1, ‘’, ‘’, ‘’,
    QgsVectorLayerDirector.DirectionBoth
)
director.addStrategy(QgsNetworkDistanceStrategy())

builder    = QgsGraphBuilder(layer.crs(), True, SNAP_TOL-
ERANCE)
snapped_pts = director.makeGraph(builder, settlement_cen-
troids)
graph      = builder.graph()

print(f”Graph built: {graph.vertexCount()} vertices, {graph.
edgeCount()} edges”)

# Cache all graph vertices once — used repeatedly below
all_graph_vertices = [(vid, graph.vertex(vid).point()) for vid in
range(graph.vertexCount())]
# ------------------------------
# FIND ONE REPRESENTATIVE ROAD VERTEX PER
CENTRALITY POLYGON
# Dijkstra is run FROM each of these vertices outward, so the
shortest
# distance from any settlement back TO a centrality is captured
in one pass.
# ------------------------------
print(“Finding representative road vertex per centrality
polygon...”)

centrality_vertex_ids = set()
for cg in centrality_geoms:
    centroid  = cg.centroid().asPoint()
    cg_bbox   = cg.boundingBox()
    best_vid  = -1
    best_dist = float(‘inf’)
    for vid, pt in all_graph_vertices:
        if not cg_bbox.contains(pt):
            continue
        if cg.contains(QgsGeometry.fromPointXY(pt)):
            d = centroid.distance(pt)
            if d < best_dist:
                best_dist = d
                best_vid  = vid
    if best_vid != -1:
        centrality_vertex_ids.add(best_vid)

print(f”Representative centrality vertices found: {len(centrali-
ty_vertex_ids)}”)

if len(centrality_vertex_ids) == 0:
    raise Exception(
        “No road vertices found inside centrality polygons. “
        “Check that the road layer and centrality layer share the
same CRS.”
    )

# ------------------------------
# REVERSE DIJKSTRA FROM EACH CENTRALITY
VERTEX
# costs_from_centrality[v] = shortest road distance from vertex
v to ANY centrality
# Running a handful of Dijkstra passes (one per centrality
vertex) is far faster
# than running one pass per settlement.
# ------------------------------
print(“Running Dijkstra from centrality vertices...”)

costs_from_centrality = [float(‘inf’)] * graph.vertexCount()

for i, cvid in enumerate(centrality_vertex_ids):
    result_a, result_b = QgsGraphAnalyzer.dijkstra(graph, cvid,
0)

    # Detect which array is costs vs tree:
    # cost arrays contain float distances; tree arrays contain
integer edge indices
    a_has_floats = any(v != int(v) for v in result_a[:50] if v >= 0)
    costs = result_a if a_has_floats else result_b

    for vid in range(graph.vertexCount()):
        if costs[vid] < costs_from_centrality[vid]:
            costs_from_centrality[vid] = costs[vid]

    print(f”  Dijkstra {i + 1}/{len(centrality_vertex_ids)} done”)
# ------------------------------
# MAP EACH SETTLEMENT TO ITS NEAREST
GRAPH VERTEX (via snapped point)
# Then look up that vertex’s cost
# ------------------------------
print(“Mapping settlements to graph vertices and reading
distances...”)

road_distances = {}

# Settlements inside a centrality  distance 0
for fid in overlapping_ids:
    road_distances[fid] = 0.0

# Remaining settlements  nearest graph vertex to their snapped
point
for f, sp in zip(non_overlapping, snapped_pts):
    best_vid  = -1
    best_dist = float(‘inf’)
    for vid, pt in all_graph_vertices:
        d = sp.distance(pt)
        if d < best_dist:
            best_dist = d
            best_vid  = vid
        if best_dist == 0.0:
            break  # exact match — no need to continue

    if best_vid == -1:
        name = str(f[‘tatort’]) if ‘tatort’ in [fi.name() for fi in
f.fields()] else str(f.id())
        print(f”  WARNING: ‘{name}’ — no graph vertex found,
using dist=0”)
        road_distances[f.id()] = 0.0
    else:
        cost = costs_from_centrality[best_vid]
        if cost < 0 or cost == float(‘inf’):
            name = str(f[‘tatort’]) if ‘tatort’ in [fi.name() for fi in
f.fields()] else str(f.id())
            print(f”  WARNING: ‘{name}’ — unreachable (cost={f-
cost}), using dist=0”)
            road_distances[f.id()] = 0.0
        else:
            road_distances[f.id()] = cost

# ------------------------------
# ISLAND / FERRY OVERRIDES
# Replace road_dist for settlements in ISLAND_SETTLE-
MENTS with the
# maximum reachable distance so they score as maximally
remote.
# ------------------------------
if ISLAND_SETTLEMENTS:
    max_reachable = max(d for d in road_distances.values() if
d > 0)
    name_to_fid  = {}
    if ‘Name’ in [fi.name() for fi in layer.fields()]:
        name_to_fid = {str(f[‘Name’]): f.id() for f in layer.getFea-
tures()}
    for island_name in ISLAND_SETTLEMENTS:
        fid = name_to_fid.get(island_name)
        if fid is not None:
            road_distances[fid] = max_reachable
            print(f”  Island override: ‘{island_name}’  {max_reacha-
ble:.0f}m (max reachable dist)”)
        else:
            print(f”  Island override WARNING: ‘{island_name}’
not found in layer”)
# ------------------------------
# DIAGNOSTIC SUMMARY
# ------------------------------
dist_vals  = list(road_distances.values())
zero_count = sum(1 for v in dist_vals if v == 0.0)
print(f”\nDistance summary:”)
print(f”  min={min(dist_vals):.0f}m  max={max(dist_vals):.0f}m
zeros={zero_count}/{len(dist_vals)}”)

fid_to_name  = {}
name_field   = ‘Name’
if name_field in [fi.name() for fi in layer.fields()]:
    fid_to_name = {f.id(): str(f[name_field]) for f in layer.
getFeatures()}

sorted_dists = sorted(road_distances.items(), key=lambda x:
x[1])
print(“\n5 closest settlements to a centrality:”)
for fid, d in sorted_dists[:5]:
    print(f”  {fid_to_name.get(fid, fid)}: {d:.0f}m”)
print(“5 furthest settlements from a centrality:”)
for fid, d in sorted_dists[-5:]:
    print(f”  {fid_to_name.get(fid, fid)}: {d:.0f}m”)

# ------------------------------
# WRITE road_dist BACK TO LAYER  (always overwrites)
# ------------------------------
print(“\nWriting road_dist to settlements layer...”)

layer.startEditing()
for f in layer.getFeatures():
    layer.changeAttributeValue(f.id(), idx_road_dist, road_dis-
tances.get(f.id(), 0.0))
layer.commitChanges()
layer.updateFields()

print(“Done. ‘road_dist’ field is now populated and ready to use
in the allocation script.”)

from qgis.core import (
    QgsProject,
    QgsVectorLayer,
    QgsField,
    QgsFeature,
    QgsGeometry,
    QgsPointXY,
    QgsVectorFileWriter,
    QgsFields,
)
from qgis.analysis import (
    QgsVectorLayerDirector,
    QgsNetworkDistanceStrategy,
    QgsGraphBuilder,
    QgsGraphAnalyzer,
)
from qgis.PyQt.QtCore import QVariant
from collections import Counter
import random
# ------------------------------
# PARAMETERS  — edit these
# ------------------------------
settlements_layer_name = “Settlements all”
road_layer_name        = “Roads”

population_field = “bef”
road_dist_field  = “road_dist”  # written by compute_road_dis-
tances.py

total_points = 13

# Population category thresholds and MAX CAPACITY per
settlement.
# A settlement can never receive more points than its capacity
across all passes.
# Settlements below the lowest threshold are ineligible (capacity
0).
POP_CATEGORIES = [
    (50,   200,   1),  # small   — max 1 point
    (200,  2500,  1),  # medium  — max 1 points
    (2500, 15000, 2),  # large   — max 2 points
    (15000, 100000, 5),  # city   — max 40% points
]

# Controls how many points Umeå gets upfront:
#   anchor_points = round(weight_pop * total_points)
# e.g. weight_pop=0.6, total_points=20  Umeå gets 12 points
# Also used as population component weight in combined
scores.
weight_pop  = 0.5
weight_dist = 0.5   # must sum to 1.0 with weight_pop

# Minimum road network distance (metres) between points in
DIFFERENT settlements.
# Enforced on every pass — a settlement blocked by a neigh-
bour in pass 1
# stays blocked in pass 2, 3 etc.
MIN_DIST_M = 15000

# Anchor settlement — gets its share upfront, exempt from
spacing constraint.
ANCHOR_SETTLEMENT = “Umeå”
ANCHOR_FIELD      = “Name”

output_path = “/tmp/redistributed_points.shp”
# ------------------------------
# GET LAYERS
# ------------------------------
layer     = QgsProject.instance().mapLayersByName(settle-
ments_layer_name)[0]
road_layer = QgsProject.instance().mapLayersByName(road_
layer_name)[0]

if road_dist_field not in [f.name() for f in layer.fields()]:
    raise Exception(
        f”Field ‘{road_dist_field}’ not found. Run compute_road_
distances.py first.”
    )
# ------------------------------
# ADD WORKING FIELDS IF MISSING
# ------------------------------
existing_fields = [f.name() for f in layer.fields()]
new_fields = []

for fname, ftype in [
    (“pop_cat”,    QVariant.Int),
    (“pop_score”,  QVariant.Double),
    (“n_dist”,     QVariant.Double),
    (“score”,      QVariant.Double),
    (“alloc_fin”,  QVariant.Int),
]:
    if fname not in existing_fields:
        new_fields.append(QgsField(fname, ftype))

if new_fields:
    layer.dataProvider().addAttributes(new_fields)
    layer.updateFields()
# ------------------------------
# FIELD INDICES
# ------------------------------
idx_pop      = layer.fields().indexOf(population_field)
idx_rd       = layer.fields().indexOf(road_dist_field)
idx_pop_cat  = layer.fields().indexOf(“pop_cat”)
idx_pop_score = layer.fields().indexOf(“pop_score”)
idx_n_dist   = layer.fields().indexOf(“n_dist”)
idx_score    = layer.fields().indexOf(“score”)
idx_alloc    = layer.fields().indexOf(“alloc_fin”)

for name, idx in [
    (population_field, idx_pop),
    (road_dist_field,  idx_rd),
    (“pop_cat”,        idx_pop_cat),
    (“pop_score”,      idx_pop_score),
    (“n_dist”,         idx_n_dist),
    (“score”,          idx_score),
    (“alloc_fin”,      idx_alloc),
]:
    if idx == -1:
        raise Exception(f”Field ‘{name}’ not found.”)
# ------------------------------
# READ VALUES
# ------------------------------
features_list = list(layer.getFeatures())
pop_values    = [f[idx_pop] or 0 for f in features_list]
dist_values   = [f[idx_rd]  or 0 for f in features_list]

print(f”Features loaded: {len(features_list)}”)
# ------------------------------
# POPULATION CATEGORIES AND CAPACITIES
# ------------------------------
cat_names = {0: “below threshold”, 1: “small”, 2: “medium”, 3:
“large”, 4: “city”}

def pop_category(pop):
    for cat_idx, (lo, hi, capacity) in enumerate(POP_CATE-
GORIES):
        if lo <= pop < hi:
            return cat_idx + 1, capacity
    return 0, 0

pop_cats     = []
pop_capacities = []
pop_scores     = []

for pop in pop_values:
    cat, capacity = pop_category(pop)
    pop_cats.append(cat)
    pop_capacities.append(capacity)
    pop_scores.append(capacity)

max_capacity  = max(c[2] for c in POP_CATEGORIES)
n_pop_scores  = [s / max_capacity for s in pop_scores]

dist_lo     = min(dist_values)
dist_rng    = (max(dist_values) - dist_lo) or 1
n_dist_values = [(d - dist_lo) / dist_rng for d in dist_values]

scores = [
    weight_pop * n_pop_scores[i] + weight_dist * n_dist_val-
ues[i]
    if pop_cats[i] > 0 else 0.0
    for i in range(len(features_list))
]

cat_counts = Counter(pop_cats)
print(“\nPopulation category breakdown:”)
for cat in sorted(cat_counts):
    print(f”  {cat_names.get(cat, cat)}: {cat_counts[cat]} settle-
ments”)
# ------------------------------
# IDENTIFY ANCHOR
# ------------------------------
anchor_indices = set()
if ANCHOR_SETTLEMENT and ANCHOR_FIELD in [fi.
name() for fi in layer.fields()]:
    for i, f in enumerate(features_list):
        if str(f[ANCHOR_FIELD]) == ANCHOR_SETTLE-
MENT:
            anchor_indices.add(i)

anchor_capacity = sum(pop_capacities[i] for i in anchor_in-
dices)
print(f”\nAnchor ‘{ANCHOR_SETTLEMENT}’: {len(an-
chor_indices)} polygon(s), “
      f”category capacity = {anchor_capacity}”)
# ------------------------------
# STEP 1 — ANCHOR ALLOCATION (upfront, exempt
from spacing)
# ------------------------------
alloc_counts = [0] * len(features_list)

if anchor_indices:
    anchor_alloc = round(weight_pop * total_points)
    anchor_alloc = min(anchor_alloc, anchor_capacity, to-
tal_points - 1)
    anchor_alloc = max(anchor_alloc, 1 if anchor_capacity > 0
else 0)
    for i in anchor_indices:
        alloc_counts[i] = anchor_alloc
    points_placed = sum(alloc_counts)
    print(f”Anchor allocation: {anchor_alloc} points to ‘{AN-
CHOR_SETTLEMENT}’”)
else:
    points_placed = 0

remaining_points = total_points - points_placed
print(f”Remaining points for multi-pass spatial distribution:
{remaining_points}”)
# ------------------------------
# STEP 2 — BUILD ROAD NETWORK FOR SPACING
# ------------------------------
print(“\nBuilding road network for spacing constraint...”)

centroids   = [f.geometry().centroid().asPoint() for f in fea-
tures_list]
director    = QgsVectorLayerDirector(
    road_layer, -1, ‘’, ‘’, ‘’,
    QgsVectorLayerDirector.DirectionBoth
)
director.addStrategy(QgsNetworkDistanceStrategy())
builder     = QgsGraphBuilder(layer.crs(), True, 10.0)
snapped_pts = director.makeGraph(builder, centroids)
graph       = builder.graph()

print(f”Graph: {graph.vertexCount()} vertices, {graph.edge-
Count()} edges”)

all_verts = [(vid, graph.vertex(vid).point()) for vid in range(-
graph.vertexCount())]

def nearest_vertex(pt):
    best_vid, best_d = -1, float(‘inf’)
    for vid, vpt in all_verts:
        d = pt.distance(vpt)
        if d < best_d:
            best_d, best_vid = d, vid
        if best_d == 0.0:
            break
    return best_vid

settlement_vids = [nearest_vertex(sp) for sp in snapped_pts]
dijkstra_cache  = {}

def get_road_dist(vid_a, vid_b):
    if vid_a == vid_b:
        return 0.0
    if vid_a not in dijkstra_cache:
        ra, rb = QgsGraphAnalyzer.dijkstra(graph, vid_a, 0)
        a_has_floats = any(v != int(v) for v in ra[:50] if v >= 0)
        dijkstra_cache[vid_a] = ra if a_has_floats else rb
    c = dijkstra_cache[vid_a][vid_b]
    return c if c >= 0 else float(‘inf’)
# ------------------------------
# STEP 3 — MULTI-PASS GREEDY ALLOCATION
#
# Each pass places at most 1 point per settlement, in score order,
# respecting MIN_DIST_M spacing. Blocked settlements stay
blocked
# across all passes. A settlement is skipped once it reaches
capacity.
# Passes repeat until all remaining_points are placed or no
eligible
# settlements remain.
# ------------------------------
print(“\nMulti-pass greedy allocation...”)

# Eligible non-anchor settlements sorted by score descending
# This order is fixed — higher scoring settlements always get
priority
non_anchor_order = sorted(
    [i for i in range(len(features_list))
     if i not in anchor_indices and scores[i] > 0],
    key=lambda i: scores[i],
    reverse=True
)

# Blocked = too close to an already-placed point (persists across
passes)
blocked = [False] * len(features_list)

# Pre-block settlements too close to anchor
for i in anchor_indices:
    for j in non_anchor_order:
        if not blocked[j]:
            if get_road_dist(settlement_vids[i], settlement_vids[j])
<= MIN_DIST_M:
                blocked[j] = True

points_placed_greedy = 0
pass_num = 0

while points_placed_greedy < remaining_points:
    pass_num += 1
    placed_this_pass = 0

    for i in non_anchor_order:
        if points_placed_greedy >= remaining_points:
            break

        # Skip if blocked by spacing or already at capacity
        if blocked[i]:
            continue
        if alloc_counts[i] >= pop_capacities[i]:
            continue

        # Place one point
        alloc_counts[i]    += 1
        points_placed_greedy += 1
        placed_this_pass   += 1

        f   = features_list[i]
        name = str(f[ANCHOR_FIELD]) if ANCHOR_FIELD
in [fi.name() for fi in f.fields()] else str(f.id())
        print(f”  Pass {pass_num} | {name} [{cat_names.get(pop_
cats[i])}]  “
              f”total={alloc_counts[i]}  score={scores[i]:.3f}  “
              f”pop={pop_values[i]}  dist={dist_values[i]:.0f}m”)

        # Block neighbours (only on first point placed at this
settlement)
        if alloc_counts[i] == 1:
            vid_i = settlement_vids[i]
            for j in non_anchor_order:
                if j != i and not blocked[j]:
                    if get_road_dist(vid_i, settlement_vids[j]) <=
MIN_DIST_M:
                        blocked[j] = True

    if placed_this_pass == 0:
        print(f”\nWARNING: No points placed in pass {pass_
num}.”)
        print(f”  All eligible settlements are blocked or at capac-
ity.”)
        print(f”  Placed {points_placed_greedy}/{remaining_points}
greedy points.”)
        print(f”  Options: lower MIN_DIST_M ({MIN_DIST_M}
m) or increase capacities.”)
        break

total_placed = sum(alloc_counts)
print(f”\nTotal points placed: {total_placed} in {pass_num}
passes”)
print(“\nFinal allocation:”)
for i, f in enumerate(features_list):
    if alloc_counts[i] > 0:
        name = str(f[ANCHOR_FIELD]) if ANCHOR_FIELD
in [fi.name() for fi in f.fields()] else str(f.id())
        print(f”  {name} [{cat_names.get(pop_cats[i])}]: {al-
loc_counts[i]} pts “
              f”(capacity={pop_capacities[i]}, score={scores[i]:.3f})”)
# ------------------------------
# WRITE BACK TO LAYER
# ------------------------------
layer.startEditing()

for i, f in enumerate(features_list):
    layer.changeAttributeValue(f.id(), idx_pop_cat,  pop_cats[i])
    layer.changeAttributeValue(f.id(), idx_pop_score, pop_
scores[i])
    layer.changeAttributeValue(f.id(), idx_n_dist,   n_dist_val-
ues[i])
    layer.changeAttributeValue(f.id(), idx_score,    scores[i])
    layer.changeAttributeValue(f.id(), idx_alloc,    alloc_
counts[i])

layer.commitChanges()
layer.updateFields()
# ------------------------------
# CREATE OUTPUT POINT LAYER
# ------------------------------
crs = layer.crs().authid()

fields = QgsFields()
fields.append(QgsField(“parent_id”, QVariant.Int))

points_layer = QgsVectorLayer(f”Point?crs={crs}”, “redistribut-
ed_points”, “memory”)
points_layer.dataProvider().addAttributes(fields)
points_layer.updateFields()

for f in layer.getFeatures():
    geom    = f.geometry()
    n_points = max(0, int(f[idx_alloc]))

    if geom.isMultipart():
        polygon = geom.asMultiPolygon()[0][0]
    else:
        polygon = geom.asPolygon()[0]

    minx = min(p.x() for p in polygon)
    maxx = max(p.x() for p in polygon)
    miny = min(p.y() for p in polygon)
    maxy = max(p.y() for p in polygon)

    for _ in range(n_points):
        while True:
            x  = random.uniform(minx, maxx)
            y  = random.uniform(miny, maxy)
            pt = QgsPointXY(x, y)
            if geom.contains(QgsGeometry.fromPointXY(pt)):
                feat = QgsFeature()
                feat.setGeometry(QgsGeometry.fromPointXY(pt))
                feat.setAttributes([f.id()])
                points_layer.dataProvider().addFeature(feat)
                break
# ------------------------------
# ADD TO PROJECT & SAVE
# ------------------------------
QgsProject.instance().addMapLayer(points_layer)

QgsVectorFileWriter.writeAsVectorFormat(
    points_layer,
    output_path,
    “UTF-8”,
    layer.crs(),
    “ESRI Shapefile”
)

print(“Done. Redistributed points layer added to project and
saved to:”, output_path)
```